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Semantic Stability in the King James Bible: Comparing the 1611 and 1769 Editions

A Human-in-the-Loop Embedding Study of Textual Drift

Using the InterLink Research Platform

Jamison Lewis

Saorsa Industries

Contents

Abstract

This study evaluates semantic stability and textual drift between the King James Version (KJV) 1611 and KJV 1769 Bible editions using an aligned unit computational workflow developed for the InterLink Research Platform. The shared corpus consists of 66 works and 31,102 structurally aligned textual units, including 23,145 Old Testament units and 7,957 New Testament units. Each aligned pair was embedded using *mxbai-embed-large*, and semantic drift was measured via cosine distance across raw text and lightly normalized representation layers. The active representations deployed in this study, namely raw text (R0) and lightly normalized text (R1), produced identical drift values across all aligned units, indicating model invariance to basic normalization operations.

Across the shared corpus, median drift was low at 0.0592 with a Central Semantic Stability Index (CSSI) of 0.9408, indicating high central semantic stability between editions. Robust intratextual thresholds identified a small upper tail population of 1,238 review candidates and 359 strong outliers, together representing 5.13% of aligned units. Monte Carlo permutation baselines showed that true aligned pairs were substantially closer than randomized pairings under intratextual, intracorporal, and exploratory supracanonical null models. Human in the loop category enrichment analysis revealed that the drift tail was dominated by orthographic and character form modernization, with the *orthographic_character_form* categorical change emerging as the only statistically significant overrepresented category after false discovery rate correction.

These findings confirm that the KJV 1611 and KJV 1769 editions are highly stable semantically at the aligned unit level under the selected embedding model, while demonstrating that modern embedding models register early modern orthographic variation as measurable semantic distance. The study validates the InterLink workflow as a foundation for computational theological text comparison and motivates future research regarding aggressive historical orthographic normalization, KJV 1611 exclusive corpus analysis, and domain specific theological embedding models.

Introduction

The relationship between historical biblical editions is frequently examined through the lens of textual variation, editorial transmission, orthographic modernization, and interpretive continuity [1]. In the case of the King James Bible, the KJV 1611 and KJV 1769 are closely related editions, yet they are not textually identical. Differences in spelling, punctuation, capitalization, typography, and occasional wording create an ideal test case for evaluating whether modern computational methods can distinguish broad semantic stability from localized textual drift.

This study examines the KJV 1611 and KJV 1769 editions using an aligned unit semantic drift workflow developed for the InterLink Research Platform. The objective is not to argue that computational embeddings can replace philological, theological, or historical judgment. Rather, the purpose is to test whether an embedding based workflow can provide a reproducible method for measuring edition level semantic stability, identifying upper tail drift candidates, and supporting human interpretation through structured annotation.

The study is framed as a foundational method validation inquiry. It asks whether a computational pipeline can successfully ingest historically related biblical editions, align comparable textual units, generate embeddings, compute semantic distance, apply robust thresholding, evaluate randomized baselines, and incorporate human in the loop interpretation of high drift cases. This framing emphasizes that the primary contribution is methodological; the study validates a workflow for computational theological text comparison before extending the platform to broader cross corpus analysis.

Motivation

Biblical texts present distinct challenges for computational comparison. They are highly structured, deeply intertextual, historically layered, and often transmitted through editions that contain both surface level and potentially meaning bearing variation [2]. Some differences are clearly orthographic, such as early modern letterforms or spelling conventions. Other variations involve punctuation, clause structure, lexical substitution, additions, deletions, and syntactic shifts. A computational workflow must therefore do more than produce a singular similarity score, it must also provide a systematic way to distinguish central stability from tail behavior while preserving a role for human interpretation.

The KJV 1611 and KJV 1769 editions are uniquely suited for this task because they are close enough to support direct structural alignment, yet sufficiently distinct to test the sensitivity of modern embeddings to historical textual variation. A successful workflow should demonstrate that the majority of aligned units remain semantically close while still detecting a small set of units requiring closer inspection. Furthermore, it should help determine whether the detected drift tail reflects substantive textual divergence or merely the behavior of the embedding model when exposed to orthographic and character form modernization.

Research Problem

Modern embedding models are increasingly leveraged for semantic search, retrieval, clustering, and textual comparison. However, their behavior on historically inflected religious texts requires careful validation [3]. A high embedding distance between two aligned biblical units may reflect a meaningful semantic difference, but it may also reflect spelling modernization, punctuation updates, archaic letterforms, abbreviation expansion, or other surface level variations. Without statistical thresholds, baseline reference models, and human annotation, embedding distance alone remains difficult to interpret.

This study addresses the problem by combining five methodological layers:

- Structural alignment of shared KJV 1611 and KJV 1769 textual units.
- Embedding based drift measurement using cosine distance.
- Robust intratextual thresholding to isolate review candidates and strong outliers.

- Monte Carlo permutation baselines to test whether true aligned pairs are significantly closer than randomized alternatives.
- Human in the loop category enrichment analysis to interpret the textual change patterns associated with high drift units.

Together, these layers permit the investigation of three distinct dimensions: whether the editions are centrally stable, whether a measurable drift tail exists, and what specific textual changes characterize that drift tail. Figure 1 provides a schematic overview of the full InterLink drift analysis workflow, beginning with corpus ingestion and structural alignment, proceeding through representation-specific embedding generation, drift computation, thresholding, permutation testing, and HITL annotation, and ending with interpretive synthesis and reproducible output artifacts.

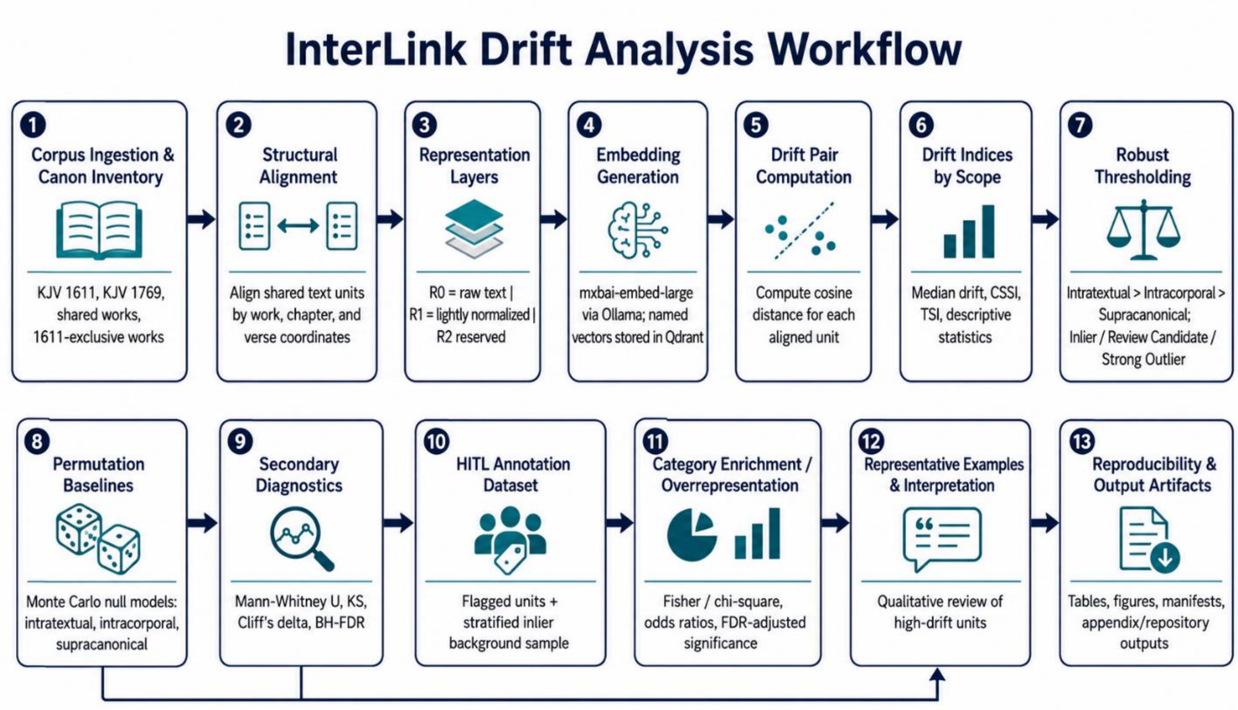


Figure 1. InterLink drift analysis workflow

Research Questions and Hypotheses

The study is organized around four core research questions:

- **RQ1**: Are true aligned KJV-1611/KJV-1769 units closer in embedding space than randomized pairings from the same textual scopes? This question tests whether structural alignment captures a meaningful semantic relationship beyond general corpus similarity.
- **RQ2**: Does the shared corpus contain a measurable upper tail population of higher drift units? This question tests whether robust thresholds can identify units that warrant interpretive review while preserving the expectation that most aligned units remain stable.

- **RQ3:** Do human annotated textual change categories show structure among flagged drift units? This question tests whether high drift units are associated with identifiable categories such as orthographic modernization, punctuation clause relation changes, lexical substitution, function word changes, pronoun shifts, negation or modality changes, syntactic reordering, additions, deletions, or complex multi factor variation.
- **RQ4:** How should KJV 1611 exclusive material be handled in the broader InterLink workflow? Because the 1611 exclusive corpus lacks direct KJV 1769 counterparts, this material must be handled outside of direct aligned drift calculations.

The corresponding hypotheses are as follows:

- **H1:** True aligned pairs will display lower semantic distance than randomized baseline pairs.
- **H2:** Robust thresholds will identify a small but meaningful upper tail population of review candidates and strong outliers.
- **H3:** Flagged units will exhibit clear category structure under human annotation.
- **H4:** KJV 1611 exclusive material is inventoried against shared canon groupings and is deferred to future nearest neighbor analysis.

Study Contribution

This study contributes a reproducible computational workflow for evaluating semantic stability between historically related biblical editions. By integrating embedding-based measurement with robust thresholds, randomized null models, and human in the loop annotation, the workflow establishes a reliable framework for text critical analysis.

Methodologically, the study demonstrates that historical orthographic variation can meaningfully affect modern embedding distances even when the underlying meaning remains largely stable [4]. Substantively, the results confirm that the shared KJV-1611/KJV-1769 corpus is centrally stable in semantic embedding space. Despite thousands of orthographic, spelling, and minor textual updates the analysis found no evidence of broad doctrinal semantic divergence at the aligned unit level. This dual insight motivates future work regarding more aggressive historical normalization and domain specific embedding models trained or adapted specifically for biblical and theological corpora.

Study Scope

The direct drift analysis is comprised of the 66 shared works between the KJV 1611 and KJV 1769. The aligned shared corpus contains 31,102 textual units, all of which have complete vector coverage for the active representation layers (R0, R1). The KJV 1611 exclusive corpus contains 15 works (treating the *Letter of Jeremiah* as separate from *Baruch*) and 5,705 units. These exclusive works are inventoried but excluded from direct aligned pair drift calculations as they lack corresponding KJV 1769 shared canon counterparts.

The study utilizes the mxbai-embed-large embedding model and cosine distance as the primary drift metric. Two active representation layers were evaluated: raw text (R0) and lightly normalized text (R1), where R1 applies lowercasing and whitespace normalization. Because R0 and R1 produced identical drift values, R0 serves as the representative layer for primary reporting, while R1 equivalence is reported as a model invariance finding. A more aggressive experimental normalization layer (R2) is disabled for this study and reserved for future work.

The remainder of the paper proceeds as follows. The Methods section describes corpus construction, representation layers, embedding generation, drift measurement, thresholding, permutation baselines, and human in the loop annotation. The Results section reports corpus coverage, drift indices, representation equivalence under character normalization, threshold classifications, permutation results, category enrichment and overrepresentation findings, and representative examples. The Discussion section interprets the findings in relation to semantic stability, orthographic modernization, embedding model behavior, and future InterLink research directions.

Methods

Study Design

This study deploys a human in the loop computational architecture to systematically evaluate semantic stability and textual drift between the KJV 1611 and KJV 1769 editions of the Bible. Serving as a foundational method validation inquiry for the InterLink Research Platform, the design establishes an empirical workflow to measure central semantic consistency while isolating and interpreting localized textual divergence. The investigative framework integrates five primary sequential layers:

- Structural corpus alignment and inventory of textual units.
- Representation specific embedding generation across raw and character normalized layers.
- Distance measurement utilizing cosine distance to quantify pairwise textual drift.
- Robust hierarchical thresholding to isolate bounded upper tail populations of high drift units.
- Monte Carlo permutation baselines and secondary distributional diagnostics to validate alignment significance against random expectations.

To preserve the interpretive integrity of the textual analysis, these quantitative metrics are coupled with human annotation of textual change categories within the drift tail. The direct comparison is strictly limited to the 66 works shared between the KJV 1611 and KJV 1769 canonical editions. Extracanonical or KJV 1611 exclusive works are inventoried independently as a separate structural baseline and are excluded from direct aligned pair drift calculations due to the absence of corresponding KJV 1769 counterparts.

Terminology for Scope Levels

To evaluate semantic stability and textual drift across multiple nested structural scales, this study establishes a consistent vocabulary to define analytical boundaries:

- **Intratextual:** This scope restricts observations and calculations to the boundaries of a single biblical work. For example, an intratextual null model for Genesis draws randomized comparison units exclusively from the text of Genesis. This represents the most granular and localized scope deployed in the study.
- **Intracorporal:** This scope expands the analytical boundary to a major corpus division. In the present study, the primary intracorporal scopes are the Old Testament shared corpus and the New Testament shared corpus.
- **Supracanonical:** This scope spans the entirety of the shared KJV-1611/KJV-1769 dataset included in the direct drift analysis. Supracanonical results provide global, corpus wide baseline context, but they serve as exploratory reference points rather than the primary inferential foundation when more localized estimates are available.

These terms correspond to work level, work-group level, and canon level analysis, respectively. By utilizing this precise structural hierarchy, the workflow explicitly avoids flattening the distinct literary styles, vocabularies, and historical transmission profiles of individual books into a single global average. Instead, this multi-tiered framework ensures that local deviations are detected and evaluated relative to their appropriate textual environments.

Text Unit Definition

For this study, a text unit refers to the smallest stable, referenceable passage used for computational comparison within a biblical canon. In the InterLink data model, each canon is represented as an ordered collection of works, and each work is divided into structured textual units identified by canonical coordinates such as work code, chapter, and verse level division. Thus, a unit such as Genesis 1:1 in KJV 1611 and Genesis 1:1 in KJV 1769 represents the same structural location within two different edition level canons. Alignment was performed by matching these shared structural coordinates across the two canons: the KJV 1611 unit served as the left-side text unit, and the corresponding KJV 1769 unit served as the right-side text unit. This approach allowed the study to compare semantically corresponding passages directly while preserving each edition's original textual form.

Corpus Construction and Scope

The study corpus comprises two distinct edition level textual canons: KJV 1611 and KJV 1769. Each canon is mapped using a structured data schema that encapsulates canon metadata, work specific attributes, and individual textual units. To ensure relational integrity across the workflow, each textual unit is mapped to a stable structural reference composed of a unique canon identifier, a standardized work code, and localized division level identifiers. The study corpora consisted of:

- **The Shared Corpus:** This core group encompasses the 66 canonical works present across both historic editions, totaling 31,102 aligned textual units. This shared architecture is divided into two operational intracorporal groups:
    - **Old Testament Shared Corpus:** Encompasses 39 biblical works and contains 23,145 aligned textual units.

- **New Testament Shared Corpus:** Encompasses 27 biblical works and contains 7,957 aligned textual units.

- **The KJV 1611 exclusive corpus:** This baseline group accounts for the 15 additional books (the apocryphal literature, separating the *Letter of Jeremiah* from *Baruch*) containing 5,705 independent textual units. These texts are inventoried as a separate isolated dataset.

Because the exclusive material lacks direct KJV 1769 counterparts within the shared canonical framework, these units are withheld from direct aligned pair drift calculations. Instead, this distinct baseline is preserved intact for future exploratory nearest neighbor and cross corpus positioning analysis.

Structural Alignment

Structural alignment was executed at the individual textual unit level. Each KJV 1611 unit within the shared corpus was systematically paired with its corresponding KJV 1769 counterpart using the unique work codes and division level indices established during corpus ingestion. This configuration maps the data into exactly one aligned pair per shared structural reference point, ensuring a balanced matrix for pairwise distance computation.

The enforcement of strict alignment validity required satisfying three core relational constraints:

- **One to one correspondence:** Every shared KJV 1611 textual unit must map directly to its corresponding KJV 1769 counterpart.
- **Structural completeness:** No aligned shared unit could be missing or omitted from either side of the paired dataset.
- **Vector continuity:** All active representation layers must maintain complete, uninterrupted vector coverage across the entire corpus profile.

Units originating from the KJV 1611 exclusive corpus were automatically filtered out from direct alignment configurations due to the structural absence of corresponding KJV 1769 shared canon records for those specific historical books.

Representation Layers

The computational pipeline evaluated two active representation layers to isolate the impact of surface level text variations on vector behavior:

- **Raw Representation (R0):** This layer utilizes the unaltered textual form of each unit. It explicitly preserves the native spelling, punctuation, capitalization, historic ampersands, and other character level features present in the source text.
- **Lightly Normalized Representation (R1):** This layer applies uniform lowercasing and whitespace normalization. It functions as an operational control to evaluate whether basic preprocessing operations alter embedding based distance metrics.

A third experimental tier, the **Aggressive Normalized Representation (R2)**, was disabled for this phase of the study and reserved intact for future iterations. This forthcoming layer is architected to test deep historical orthographic normalization and automated lexical mapping.

Following vectorization and distance computation, R0 and R1 yielded identical drift values across all 31,102 aligned units. Consequently, R0 serves as the primary representative layer throughout all reporting and synthesis. The observed numerical equivalence between layers is treated explicitly as a model invariance finding under the selected architecture, rather than an independent confirmation or replication of the structural drift trends.

Embedding Generation

Vector embeddings were generated at the individual textual unit level using the mxbai-embed-large model deployed through the Ollama execution framework. The pipeline processed each unit independently for all active representation layers, storing and retrieving the resulting high dimensional vectors using dedicated, named vector fields mapped directly to their corresponding representation tier stored in a Qdrant vector database collection.

To allow for reproducibility and data integrity, the embedding workflow tracked a rigorous provenance schema for each computational artifact:

- Representation parameters, identifying the specific text processing layer.
- Model identifiers, confirming the embedding model used to produce the named vectors stored in the vector database.
- Stable unit identifiers, tying vectors directly back to the canonical database index.
- Cryptographic checksums, verifying the static integrity of the underlying text sequence.

This persistent metadata tracking enabled automated coverage validation prior to statistical analysis. Consequently, the downstream drift calculations operate on the intended model architecture and representation layers, and not stale, absent, or mismatched vector representations.

Semantic Drift Metric

To quantify textual divergence, semantic drift is modeled as the geometric distance between corresponding vector representations in a shared embedding space. For each aligned textual unit, drift is calculated as the cosine distance between the KJV 1611 embedding vector and its KJV 1769 counterpart. Under this metric, a lower distance signifies greater semantic similarity and vector alignment, whereas a higher distance represents greater measured drift.

High dimensional embedding vectors are automatically unit normalized upon ingestion into the Qdrant vector database. As a result, the inner product of the two vectors is equivalent to their cosine similarity. Therefore, for each aligned unit $\boldsymbol{i}$, the pairwise semantic drift $\boldsymbol{d_i}$ is computed using the simplified dot product formula:

$$d_i = 1 - (\boldsymbol{v}_i^{1611} \cdot \boldsymbol{v}_i^{1769})$$

This mathematical operation yields exactly one scalar drift value per aligned structural unit for each active representation:

$$D_{raw} = \{d_i^{R0}, d_{i+1}^{R0}, \dots, d_n^{R0}\}$$
$$D_{norm} = \{d_i^{R1}, d_{i+1}^{R1}, \dots, d_n^{R1}\}$$

where $\boldsymbol{D}_{\mathbf{raw}}$ and $\boldsymbol{D}_{\mathbf{norm}}$ represent the sets of drift values calculated from the two different active text representations (e.g., raw text and normalized text), $\boldsymbol{d}$ denotes an individual drift value within the respective set, $\boldsymbol{R0}$ and $\boldsymbol{R1}$ indicate the specific text representation method used to compute the drift, $\boldsymbol{i}$ is the starting index of the text segment being compared, $\boldsymbol{n}$ is the total number of items or the final index in the sequence.

The resulting distribution is evaluated using a suite of descriptive statistics, including the mean, standard deviation, minimum, maximum, and targeted upper percentiles. The median is explicitly selected as the primary indicator of central semantic stability due to its resistance to upper tail outliers. Concurrently, the 95th percentile and 99th percentile boundaries are deployed to isolate and characterize extreme tail behavior within the historical corpus.

Central and Tail Semantic Stability Indices

To transform geometric drift metrics into interpretable, normalized benchmarks of continuity, this study introduces two complementary frameworks: the Central Semantic Stability Index (CSSI) and the Tail Semantic Stability Index (TSI).

Central Semantic Stability Index (CSSI)

The CSSI quantifies the typical or central semantic stability observed across aligned textual units within a given corpus scope $\boldsymbol{S}$ and representation layer. It is defined as:

$$CSSI_S^{R0} = 1 - \text{median}(D_S^{R0})$$
$$CSSI_S^{R1} = 1 - \text{median}(D_S^{R1})$$

where $\boldsymbol{D}$ represents the vector of computed pairwise cosine distances. Because the underlying metric is derived from cosine distance, the CSSI is directly interpretable as the median cosine similarity between true aligned units under the selected embedding architecture. Elevated CSSI values indicate strong central semantic conservation between historical editions.

Tail Semantic Stability Index (TSI)

To evaluate the behavior of the most volatile regions of the dataset, the TSI measures stability at high percentile boundaries of the drift distribution, specifically the 95th and 99th percentiles. These benchmarks isolate worst case semantic divergence and are defined as:

$$TSI95_S^{R0} = 1 - P_{95}(D_S^{R0})$$
$$TSI95_S^{R1} = 1 - P_{95}(D_S^{R1})$$

$$TSI99_S^{R0} = 1 - P_{99}(D_S^{R0})$$
$$TSI99_S^{R1} = 1 - P_{99}(D_S^{R1})$$

Where $\boldsymbol{P_{95}}$ and $\boldsymbol{P_{99}}$ denote the 95th and 99th percentiles of the distance distribution, respectively. These indices complement the CSSI by exposing whether the highest drift regions of the corpus maintain a reasonable degree of semantic proximity or diverge more sharply than the central distribution.

Robust Stratified Thresholding

To systematically isolate genuine semantic anomalies from baseline variation, the pipeline deploys a robust stratified thresholding framework. This statistical architecture classifies aligned textual units into three mutually exclusive categories: inliers, review candidates, and strong outliers. Threshold parameters are computed independently for each individual textual stratum and representation layer, matching boundaries directly to the unique observed drift distribution of that specific pair.

For a designated textual stratum $\boldsymbol{s}$ and representation $\boldsymbol{r}$, the complete drift distribution is defined as the set:

$$D_{\mathrm{sr}} = \{d_1, d_2, \ldots, d_{\mathrm{n}}\}$$

where the operational strata encompass intratextual scopes, intracorporal scopes, and the supracanonical scope. The active sample size metric tracking the cardinality of each subset is expressed as:

$$n_{\mathrm{sr}} = |D_{\mathrm{sr}}|$$

Dispersion and Standard Decision Rules

To quantify statistical dispersion within a stratum without suffering distortion from the extreme tail values under review, the framework utilizes the Median Absolute Deviation (MAD) because it is less sensitive to extreme upper tail values. For each stratum $\boldsymbol{s}$ and representation $\boldsymbol{r}$, MAD was defined as:

$$MAD_{sr} = median(|d_{\mathrm{i}} - median(D_{sr})|)$$

When a stratum representation pair exhibits a sufficient sample size alongside non-degenerate dispersion, the pipeline triggers the standard threshold decision rule:

$$n_{\mathrm{sr}} \geq 50 \;\; and \;\; MAD_{\mathrm{sr}} > 0$$

Under these optimal conditions, the primary threshold ($\boldsymbol{T_{sr}}$) and secondary threshold ($\boldsymbol{R_{sr}}$) introduce a dual maximum constraint, mapping mathematical boundaries as follows:

$$T_{sr} = max\left(P_{99,sr}\ , median(D_{sr}) + 3 \times MAD_{sr}\right)$$

$$R_{sr} = max\left(P_{95,sr}\ , median(D_{sr}) + 2.5 \times MAD_{sr}\right)$$

Where $\boldsymbol{P_{95}}$ and $\boldsymbol{P_{99}}$ represent the 95th and 99th percentiles of the localized distribution, respectively.

Algorithmic Safeguards and Small Sample Contingencies

To protect the pipeline against numerical instability or mathematical collapse in exceptional data environments, the framework embeds two automated safeguard operations:

1. **Small Sample Contingency Protocol:** Triggered when $\boldsymbol{n}_{\mathrm{sr}} < \mathbf{50}$. Because high percentile estimates become highly volatile in small datasets, the rule bypasses percentile maximum constraints entirely. It calculates boundaries utilizing the MAD components directly:

$$T_{sr} = median(D_{sr}) + 3 \times MAD_{sr}$$
$$R_{sr} = median(D_{sr}) + 2.5 \times MAD_{sr}$$

   This conditional framework preserves robust localized threshold assignments while insulating short works or small strata from unstable tail estimations.

2. **Low Dispersion Safeguard:** Triggered when $\boldsymbol{MAD}_{\mathrm{sr}} = \mathbf{0}$ within non-small sample strata. In highly uniform textual environments, a zero value for dispersion causes the MAD based component to collapse to the median, yielding overly permissive parameters. Under this specific boundary condition, the pipeline switches to a percentile only rule to prevent degenerate median based behavior:

$$T_{\mathrm{sr}} = P_{99,sr}$$
$$R_{\mathrm{sr}} = P_{95,sr}$$

Stratified Classification and Precedence Hierarchy

Following the stable computation of these threshold metrics, individual textual units are categorized into operational classifications based on their scalar drift values (d_i):

- **Strong Outliers:** Aligned units whose measure drift meets or exceeds the primary threshold boundary:

$$d_i \geq T_{sr}$$

- **Review Candidates:** Aligned units whose measure drift falls at or above the secondary threshold but remains below the primary outlier cap:

$$R_{sr} \leq d_i < T_{sr}$$

- **Inliers:** Aligned units whose measured drift resides safely below the secondary threshold boundary:

$$d_i < R_{sr}$$

To ensure that books with naturally higher or lower baseline drift distributions are not systematically over flagged or under flagged by a single global average, these classifications are governed by a strict local first precedence hierarchy:

$$intratextual \succ intracorporal \succ supracanonical$$

this processing sequence dictates that the pipeline first attempts to execute an intratextual classification computed at the individual work level. If a local distribution is deemed unavailable due to extreme data constraints (such as a work containing too few usable aligned pairs or exhibiting degenerate distributions that violate safeguard tolerances), the pipeline falls back to the broader intracorporal major corpus division scale. If the stratum is likewise compromised, the system defaults to the global supracanonical baseline of the full shared corpus.

This structural hierarchy determines the precise contextual reference distribution against which that distance is evaluated. In the present study, all unit level classifications were successfully completed utilizing primary intratextual thresholds. Consequently, every flagged textual element was judged strictly relative to the localized drift behavior of its own immediate literary environment.

Monte Carlo Permutation Baselines

To rigorously evaluate the significance of the observed text alignment, the methodology implements a Monte Carlo permutation framework. This procedure tests the null hypothesis that the measured semantic proximity between aligned KJV 1611 and KJV 1769 pairs is indistinguishable from random pairings within shared textual boundaries. By executing random permutations of the target text layer while holding the source layer fixed, the framework constructs empirical null distributions of randomized median distances across three nested structural scopes.

Formal Mathematical Framework and Notation

For each representation layer $r \in \{R_0, R_1\}$, let an individual aligned textual unit i within the global index set I consist of a source vector and a target vector:

$$\mathbf{L}_{i,r} = \text{KJV 1611 vector for unit } i \text{ under representation } r$$

$$\mathbf{R}_{i,r} = \text{KJV 1769 vector for unit } i \text{ under representation } r$$

All vector outputs are unit normalized upon database ingestion $\left(\|\mathbf{L}_{i,r}\| = \|\mathbf{R}_{i,r}\| = 1\right)$ thus, the observed pairwise semantic drift $d_{i,r}$ is computed using the scalar dot product equivalence:

$$d_{i,r} = 1 - \left(\mathbf{L}_{i,r} \cdot \mathbf{R}_{i,r}\right)$$

For a designated structural evaluation scope s (representing an individual work, a major corpus division, or the global shared canon), the observed drift distribution is defined as the set:

$$D_{\text{obs},s,r} = \left\{d_{i,r}: i \in s\right\}$$

The central tendency of this empirical distribution is quantified by its median, selected for its robust resistance to upper tail outliers:

$$M_{\text{obs},s,r} = \text{median}\left(D_{\text{obs},s,r}\right)$$

Null Model Constraints and Candidate Pools

The permutation routine systematically breaks the true historical alignment by replacing the authentic target unit $\mathbf{R}_{i,r}$ with a randomized counterpart drawn from a constrained valid candidate pool. The source vectors $\mathbf{L}_{i,r}$ remain statically fixed in their true sequential order. The framework evaluates three distinct null models (***m***), ordered by decreasing geographic localization:

1. **Intratextual Null Model (m = intra)**: Restricts the randomized target selection pool to the boundaries of the exact same biblical work. This configuration preserves work specific vocabulary, length distributions, and stylistic profiles:

$$C_{\text{intra}}(i,r) = \left\{\mathbf{R}_{j,r}\text{:work}(j) = \text{work}(i)\right\}$$

2. **Intracorporal Null Model (m = corp)**: Expands the selection pool to a major corpus division, specifically the full Old Testament shared corpus or the New Testament shared corpus:

$$C_{\text{corp}}(i,r) = \left\{\mathbf{R}_{j,r}\text{:corpus division}(j) = \text{corpus division}(i)\right\}$$

3. **Supracanonical Null Model (m = sup)**: Removes internal structural boundaries, permitting randomized selection across the global breadth of the shared canonical dataset:

$$C_{\text{sup}}(i, r) = \{\mathbf{R}_{j,r}: j \in I\}$$

Permutation Iteration and Null Distribution Generation

For a chosen null model m, each discrete Monte Carlo iteration k generates a randomized bijective assignment mapping function $\pi_{k,m}(i)$ such that the assigned target vector satisfies the structural scope restriction:

$$\mathbf{R}_{\pi_{k,m}(i),r} \in C_m(i, r)$$

During iteration k, the randomized pairwise drift values are recomputed across all units within the tested scope:

$$d^*_{i,r,k,m} = 1 - \left(\mathbf{L}_{i,r} \cdot \mathbf{R}_{\pi_{k,m}(i),r}\right)$$

The complete collection of these randomized distances forms the conditional iteration set:

$$D^*_{s,r,k,m} = \{d^*_{i,r,k,m}: i \in s\}$$

Each unique iteration resolves to exactly one scalar randomized median value:

$$M^*_{s,r,k,m} = \text{median}\left(D^*_{s,r,k,m}\right)$$

By repeating this sampling process over K independent iterations, the pipeline constructs the finalized empirical null distribution vector of randomized medians:

$$N_{s,r,m} = \{M^*_{s,r,1,m}, M^*_{s,r,2,m}, \dots, M^*_{s,r,K,m}\}$$

To optimize computational efficiency while maximizing statistical precision where inferential testing is most critical, the total iteration volume is stratified by null model level:

$$K_{\text{intra}} = 5000$$
$$K_{\text{corp}} = 1000$$
$$K_{\text{sup}} = 500$$

The intratextual null model receives the highest operational iteration execution count because it constitutes the primary inferential baseline for the study. The intracorporal and supracanonical configurations serve as secondary and exploratory broad corpus robustness checks.

Inferential Metrics and Probability Estimation

The true aligned median $M_{\text{obs},s,r}$ is evaluated directly against the corresponding empirical null distribution vector $N_{s,r,m}$. The directional research hypothesis states that structural text alignment minimizes vector distance, implying that the observed median should reside significantly below the center of the randomized distribution:

$$M_{\mathrm{obs},s,r} < \mathrm{median}(N_{s,r,m})$$

The exact permutation p value is estimated using an add one correction to eliminate zero probability boundaries and adjust for the finite limits of Monte Carlo sampling:

$$p_{s,r,m} = \frac{1 + \sum_{k=1}^{K} \mathbf{1}\left[M^{*}_{s,r,k,m} \leq M_{\mathrm{obs},s,r}\right]}{K+1}$$

The term $\mathbf{1}[\cdot]$ represents an indicator function returning a value of 1 when the randomized median meets or falls below the observed aligned baseline, and 0 otherwise. This formulation establishes a strict mathematical floor for the obtainable significance thresholds based entirely on iteration volume:

$$p_{\mathrm{min,intra}} = \frac{1}{5000+1} \approx 0.000200$$
$$p_{\mathrm{min,corp}} = \frac{1}{1000+1} \approx 0.000999$$
$$p_{\mathrm{min,sup}} = \frac{1}{500+1} \approx 0.001996$$

Consequently, variations in minimum reported empirical p values across the respective null layers reflect algorithmic iteration constraints rather than differences in true underlying effect magnitudes. To control the familywise error rate across multiple concurrent work level hypotheses within the intratextual scope, the Benjamini Hochberg false discovery rate procedure is applied to all raw intratextual p values.

Stability Gap Quantifications

To measure the absolute physical separation between authentic historical continuity and randomized baseline drift without dependence on sample size scaling, the framework defines the Stability Gap Index (SGI):

$$\mathrm{SGI}_{s,r,m} = \mathrm{median}(N_{s,r,m}) - M_{\mathrm{obs},s,r}$$

A positive scalar SGI confirms that randomized re pairing inflates semantic distance. To achieve a standardized effect size metric capable of uniform comparison across highly disparate literary genres, textual lengths, and baseline distance properties, the metric is converted to a proportion of the null center via the Normalized Stability Gap Index ($\mathrm{SGI}^{\mathrm{norm}}$):

$$\mathrm{SGI}^{\mathrm{norm}}_{s,r,m} = \frac{\mathrm{median}(N_{s,r,m}) - M_{\mathrm{obs},s,r}}{\mathrm{median}(N_{s,r,m})}$$

This index isolates the percentage reduction in semantic distance directly attributable to true structural text alignment, completing the formal quantitative validation framework.

Secondary Statistical Diagnostics

To determine whether the structural alignment between the textual layers alters the broader geometry of the distance fields beyond a simple shifting of the central tendency, the methodology incorporates secondary statistical diagnostics. These diagnostics serve as non-parametric robustness checks rather than primary inferential benchmarks, evaluating whether the observed and null distance distributions differ across their full range, dispersion, and stochastic properties.

Non-Parametric Distributional Comparisons

The framework implements the two-sample Mann Whitney U test (also known as the Wilcoxon rank sum test) to evaluate whether the rank order distribution of true aligned distances differs systematically from the rank order distribution of randomized null distances. Let $D_{\mathrm{obs},s,r}$ represent the set of true aligned distances for a given scope and representation layer and let $D^{*}_{\mathrm{null},s,r,m}$ represent the pooled set of randomized distances generated across all iterations of null model m. The test statistic is calculated by ranking the combined elements of both sets:

$$U = \sum_{i \in s} \mathrm{rank}\left(d_{i,r}\right) - \frac{n_{\mathrm{obs}}(n_{\mathrm{obs}} + 1)}{2}$$

where n_{obs} denotes the cardinality of the observed sample set. The directional research hypothesis specifies that true aligned pair distances are stochastically smaller than randomized null distances, corresponding to a lower average rank for the observed sample.

To assess global discrepancies in the overall shape, symmetry, and tails of the empirical cumulative distribution functions (ECDF), the two sample Kolmogorov Smirnov (KS) test is applied. The KS statistic quantifies the supremum distance between the empirical cumulative distribution of the observed aligned data, $F_{\mathrm{obs}}(x)$, and the empirical cumulative distribution of the randomized null baseline, $F_{\mathrm{null}}(x)$:

$$D_{\mathrm{KS}} = \sup_{x} \left| F_{\mathrm{obs}}(x) - F_{\mathrm{null}}(x) \right|$$

This diagnostic detects subtle changes in structural alignment behavior that are invariant to the median, such as variance contractions or tail skewness shifts induced by localized semantic stability.

Stochastic Dominance and Effect Size Quantification

To express the practical magnitude of the distributional separation without relying on sample size dependent significance thresholds, Cliff's delta (δ) is calculated as a robust, non-parametric effect size measure. Cliff's delta measures the probability that a randomly selected distance from the observed aligned distribution is greater than a randomly selected distance from the randomized null distribution, minus the reverse probability:

$$\delta = \frac{1}{n_{\text{obs}} n_{\text{null}}} \sum_{i=1}^{n_{\text{obs}}} \sum_{j=1}^{n_{\text{null}}} \text{sign}\left(d_{i,\text{obs}} - d_{j,\text{null}}\right)$$

The scalar output is bounded strictly within the interval $[-1,1]$. The directional properties of this metric map directly to the underlying research questions:

- **Positive Values ($\delta > 0$)**: Indicate that observed aligned distances tend to exceed randomized null distances.
- **Zero Value ($\delta = 0$)**: Indicates total stochastic equivalence between the true and randomized pairings.
- **Negative Values ($\delta < 0$)**: Confirm that observed aligned distances are stochastically smaller than randomized distances, meaning the authentic historical alignment systematically drives the vectors closer together in embedding space.

Large negative magnitudes provide explicit mathematical validation for the directional conclusions established by the primary permutation baseline analysis. By combining rank sum tracking, maximum cumulative distance estimation, and stochastic dominance mapping, these secondary diagnostics demonstrate that the semantic separation between true historical continuities and random baselines reflects a comprehensive distributional divergence across the entire textual corpus.

Human in the Loop Annotation Dataset

To interpret the qualitative dimensions of textual variance driving high embedding distances, a human in the loop (HITL) annotation protocol was deployed. The evaluation architecture maps a dual selection model comprising a flagged treatment population and a stratified control population. The flagged population consists of all text units categorized as review candidates or strong outliers under the robust threshold criteria established in the primary methodology. The control population, designated as the inlier background set, provides a non-flagged baseline to enable category overrepresentation modeling.

Stratified Sampling Engine Mechanics

The inlier background set is constructed via a deterministic stratified sampling engine designed to preserve structural and distributional balance across the embedding layers. The input universe for the selection pipeline is restricted strictly to the set of aligned units remaining within the expected drift boundary:

$$I_{\text{eligible}} = \left\{ i \in I : d_{i,r} < R_{sr} \wedge d_{i,r} \neq \emptyset \right\}$$

The selection engine partitions this eligible background pool by executing a multistage grouping operation across three categorical dimensions:

1. **Representation Layer (*r*)**: Isolates the distinct vector fields generated by the independent embedding configurations $R0 \ and \ R1$.

2. **Work Structural Units (w)**: Groupings restricted by individual work codes to ensure representation across all historical source documents.

3. **Drift Quantile Bins (q)**: To avoid sampling bias toward either the center or the margins of the stable text distribution, the engine maps unit drift values into quantile intervals calculated independently within each representation layer. If an individual stratum exhibits insufficient distance variance to resolve unique quantile boundaries, the interval formulation collapses automatically to a single uniform default bin.

Let $S_{r,w,q}$ represent a discrete stratum defined by the intersection of representation layer r, work code w, and drift quantile bin q. For each unique stratum, the sampling engine applies a pseudo random selection function governed by a fixed deterministic seed. The number of text units extracted from a given stratum is upper bounded by a static volumetric cap parameter, denoted as C. The cardinality of the sampled rows within a given stratum is governed by:

$$n(S_{r,w,q}) = min(|S_{r,w,q}|, C)$$

If the total number of available inlier rows within an individual stratum falls short of the volumetric cap ($|S_{r,w,q}| < C$), the engine retains the entire subpopulation of that stratum. This design implies that the global background sample size is an emergent property of the number of active representation work quantile strata, the configured per stratum ceiling, and the true density of the underlying data grid.

In the empirical execution of this study, the stratification routine yielded exactly 526 unique inlier text units, translating to 1,052 discrete evaluation rows distributed equally across the R0 and R1 layers. This framework produces a tractable, structurally proportional control baseline to verify whether specific linguistic change modalities are uniquely isolated within the drift tail or are pervasive across the stable body of the corpus.

Annotation Modalities and Protocol

Human evaluation was conducted on the raw, unmutated left-side text and right-side text of each aligned pair. Multiple labels were permitted to be assigned to each individual evaluation row. This multilabel configuration acknowledges that a single historical text alignment can exhibit concurrent linguistic shifts, such as localized spelling modernizations appearing alongside structural reordering of clauses.

To prevent false positive inflation, rows displaying total structural and character level identity were excluded from textual change consideration. Aligned units with zero observable difference were left unassigned by default, ensuring that identical text inlier rows do not serve as erroneous evidence for any category profile.

Annotation Category Schema

The taxonomic framework for the HITL protocol contains nine mutually non-exclusive linguistic change categories designed to capture morphological, syntactic, and semantic drift:

- orthographic_character_form
- punctuation_clause_relation
- lexical_substitution
- function_word_change
- pronoun_shift
- negation_modality_change
- word_order_syntactic_shift
- addition_deletion
- multi_factor_complex

Boundary Definitions and Criteria

The *orthographic_character_form* designation encompasses variation in spelling, historical character shapes, capitalization shifts, expansions or contractions of abbreviations, and character level modernizations that preserve the underlying lexical meaning and syntactic function. Typographical adjustments such as the expansion of an ampersand logogram into the full conjunction string are classified under this label rather than being categorized as structural or functional word shifts.

The *punctuation_clause_relation* category captures alterations in punctuation structures that directly impose structural variation on clause relations, scope limits, coordinates, explanation markers, clausal contrast, or syntactic attachments. A transition from a colon delimiter to a semicolon delimiter is classified under this category whenever the modification reshapes the formal logical connection between adjoining propositions.

The remaining structural classes track specific morphological and lexicographical behaviors (a full category schema with definitions and examples is provided in Table 5 in the Results section):

- *lexical_substitution* isolates variations in semantic content words while maintaining syntactic roles.
- *function_word_change* tracks adjustments in functional markers, particles, and connective terms.
- *pronoun_shift* isolates variations in pronominal reference types or grammatical person.
- *negation_modality_change* accounts for shifts in conditional, modal, or negative operators.
- *word_order_syntactic_shift* identifies alterations in linear constituent sequencing or structural syntax.
- *addition_deletion* records absolute insertions or omissions of textual content.
- *multi_factor_complex* is reserved for compounded variations where multiple overlapping linguistic changes occur simultaneously within a single unit.

HITL Category Overrepresentation Analysis

The final methodological stage calculates the statistical enrichment of each annotation category within the high distance drift tail relative to the stratified background control set. In this framework, enrichment is defined strictly as a measure of statistical overrepresentation within a targeted category rather than a feature transformation step.

Independence and Representation Constraints

Due to the vector fields under R0 and R1 generating identical pairwise distance distributions, utilizing both layers simultaneously within a standard frequentist inference design would violate the independence assumptions required for categorical testing. To maintain mathematical integrity, the R0 layer is established as the sole primary representative dataset for computing category rates and enrichment significance. The secondary R1 annotations are maintained within a separate union summary to preserve human observations, but they are strictly excluded from the inferential contingency modeling to avoid artificial sample size inflation.

Frequentist Overrepresentation Modeling

For each discrete annotation category c a two-by-two contingency matrix is constructed to contrast the frequency counts of the flagged tail population against the stratified background sample. Let a_c and b_c denote the counts of presence and absence of category c within the flagged population and let c_c and d_c represent the corresponding presence and absence counts within the background pool.

Population	**Category Present**	**Category Absent**
Flagged Tail	a_c	b_c
Background Control	c_c	d_c

The statistical significance of category overrepresentation is evaluated using frequentist categorical tests chosen based on cell density parameters. Where cell counts are sparse (a_c, b_c, c_c, $d_c < 5$), Fisher's exact test is computed directly using the hypergeometric distribution to determine the exact probability of the observed configuration under the null hypothesis of equal category rates:

$$p = \frac{\binom{a_c+b_c}{a_c}\binom{c_c+d_c}{c_c}}{\binom{a_c+b_c+c_c+d_c}{a_c+c_c}}$$

Where sample thresholds permit, standard Pearson Chi square testing is utilized. The magnitude and direction of the overrepresentation effect size are quantified via the sample odds ratio (OR_c):

$$\mathrm{OR}_c = \frac{a_c \cdot d_c}{b_c \cdot c_c}$$

An odds ratio greater than unity ($\mathrm{OR}_c > 1$) indicates that the linguistic change category is preferentially enriched in the drift tail.

Multiple Comparison Correction

Because enrichment hypotheses are tested concurrently across the entire suite of annotation categories, the framework controls the false discovery rate (FDR) across the family of tests. The raw frequentist p values from the category tests are adjusted using the linear step up Benjamini Hochberg procedure. The independent category p values are ordered sequentially such that $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$, where m represents the total number of tested categories. The adjusted significance threshold for each rank i is defined as:

$$p_{(i)} \leq \frac{i}{m} \cdot \alpha$$

where α is the designated global false discovery rate constraint (defined as 0.05). This multistage framework validates whether high vector drift values are directly tied to human interpretable linguistic transformations or reflect unstructured algorithmic variance.

Representative Example Selection

To provide a concrete qualitative anchor for the mathematical distance fields, a structured case selection protocol was established to extract illustrative text alignments. This selection mechanism isolates individual records from the finalized outlier ranking and human in the loop interpretability datasets. The primary operational objective is to demonstrate the direct interplay between numeric vector variance, robust threshold boundaries, and specific human verified categories of textual change.

Selection Criteria and Interpretability Goals

The selection pipeline prioritizes text alignments that exhibit a clear, visible relationship between high measured embedding distances and explicit linguistic modifications. This approach prevents the misinterpretation of elevated vector drift as automatic evidence of substantive semantic, doctrinal, or theological divergence. The examples serve as diagnostic archetypes illustrating how localized variations in syntax, spelling, or mechanics manifest within the vector space.

The resulting index consists of ten distinct aligned textual units chosen to reflect diversity across both the underlying structural scopes and the observed text change taxonomy. For each selected unit, the qualitative analysis table records the following structured parameters:

- **Unit Reference**: The canonical book, chapter, and verse index.
- **R0 Drift Metric**: The scalar distance computed via the unit normalized dot product of representation R0 vectors.

- **Threshold Classification**: The assigned ordinal label determined by the outlier bounds.
- **Primary Observed Change Pattern**: The full set of overlapping linguistic change markers assigned during manual evaluation.
- **Observed Text Level Changes**: Textual changes observed in the unit reference.

Reproducibility and Output Artifacts

To enable computational transparency and replication of the empirical workflow, the execution pipeline generates a standardized suite of structured data artifacts at each major operational stage. These artifacts decouple the computational logic from the underlying storage layers, providing permanent validation records for the entire statistical framework.

Pipeline Output Matrix

The algorithmic pipeline compiles and exports a comprehensive repository of structured tables, vectors, and diagnostic arrays:

- **Corpus Coverage Summaries**: Quantified logs verifying string matching completeness and token densities across the historical editions.
- **Aligned Drift Pair Tables**: The global master data frames containing computed distance metrics for all shared units.
- **Drift Indices by Scope**: Aggregated central tendency summaries partitioned by individual work, corpus division, and global canon.
- **Threshold Summaries**: The calculated upper tail boundaries derived from the percentiles and median absolute deviations.
- **Drift Labels and Outlier Rankings**: Prioritized index sets sorting all text units by descending distance values alongside their assigned classification layers.
- **Permutation Baseline Summaries**: The empirical null distributions of randomized medians compiled across the three stratified Monte Carlo configurations.
- **Statistical Diagnostics**: The complete array of rank sum scores, cumulative distribution divergences, and stochastic dominance metrics.
- **HITL Category Counts and Enrichment Results**: The contingency matrices, hypergeometric probabilities, and adjusted false discovery rate values.
- **Annotation Datasets**: The complete multi-label human evaluation records containing both the treatment and the stratified background control rows.

Supplementary Data Partitions

To preserve the readability of the core manuscript, large computational artifacts are not reproduced in full within the printed appendices. The appendices provide artifact descriptions, file names, row counts where applicable, and persistent repository pointers for each large output. The complete generated study artifacts, architecture documentation, validation notebooks, exploratory analysis notebooks, figure generation notebooks, table generation notebooks, diagnostic outputs, annotation datasets, and reproducibility materials are deposited in a public versioned research repository. The proprietary InterLink platform source code is not released as part of this study. However, the repository includes the analysis and validation materials necessary to audit the reported results, reproduce the manuscript tables and figures, and inspect the generated outputs used in the study. The cited repository release preserves the artifact state used for this paper, including file paths, release metadata, and checksums where applicable.

Appendix A contains the full intratextual drift index summary and provides the repository location for the complete machine-readable drift index table. Appendix B contains the threshold summary index and links to the complete threshold summary by scope and representation. Appendix C contains a ranked outlier summary and links to the complete outlier ranking artifact. Appendix D summarizes the permutation baseline outputs and secondary statistical diagnostics and links to the complete work level and corpus level result files. Appendix E summarizes the HITL annotation outputs and links to the complete category annotation dataset and unique unit inventory. Appendix F documents the KJV 1611 exclusive corpus inventory and links to the complete inventory artifact. Appendix G provides the reproducibility manifest, including repository release metadata, analysis material locations, artifact filenames, checksums where applicable, and execution notes. Appendix Figure A1 provides the work level drift variation visualization referenced in the Results section.

Results

Corpus Coverage and Alignment Validity

The shared KJV 1611 and KJV 1769 datasets exhibited total structural alignment across the 66 shared canonical works selected for the direct drift analysis. The complete evaluation grid comprised of 31,102 aligned textual units available for comparison, consisting of 23,145 Old Testament units and 7,957 New Testament units. Each structural reference unit matched a valid left side KJV 1611 unit to an equivalent right side KJV 1769 unit, yielding zero missing observation rows across the shared corpus.

Vector extraction achieved total coverage across both active representation layers. The raw text representation (***R0***) and lightly normalized representation (***R1***) each contained embeddings for all 31,102 aligned shared units. The experimental normalization representation (***R2***) was structurally deactivated for this study and reserved for future analysis. The unique KJV 1611 exclusive corpus, containing 15 works and 5,705 textual units, was documented as an independent text inventory but was excluded from the aligned pair drift analysis due to the lack of KJV 1769 counterfactual pairs. These baseline tracking results confirm that the empirical findings are free from alignment gaps, vector omission artifacts, or unequal sample densities across the shared corpus layers. Table 1 details the complete corpus coverage and alignment validity.

Table 1. Corpus Coverage and Alignment Validity

Corpus/Coverage Component	Works	Textual Units/Embeddings	Alignment Status	Analysis Role
Shared corpus	66	31,102	Yes	Direct aligned pair drift analysis
Old Testament shared corpus	39	23,145	Yes	Intracorporal analysis scope
New Testament shared corpus	27	7,957	Yes	Intracorporal analysis scope
KJV 1611 exclusive corpus	15	5,705	No	Inventoried separately. Excluded from direct drift calculations
R0 vector coverage	66 shared works	31,102 embeddings	Complete	Primary reporting layer
R1 vector coverage	66 shared works	31,102 embeddings	Complete	Light normalization invariance check

Note. Repository linked machine readable artifacts provide full precision values and row level outputs where applicable.

Overall Semantic Stability

The calculated pairwise distance values confirm that high central semantic stability characterizes the transition between the historical KJV 1611 and modernized KJV 1769 text layers. Across the breadth of the global shared corpus, the empirical distribution resolved to a median distance of 0.0592 and a mean distance of 0.0706. The upper tail of the distribution reached a 95th percentile threshold of 0.1718 and a 99th percentile threshold of 0.2472. This behavior indicates that while the vast majority of aligned units remained close in embedding space a measurable upper tail remains available for localized threshold based structural review.

The central tendency of the vector field synthesized by the Central Semantic Stability Index (***CSSI***), which operationalizes the global median similarity across a given representation stratum. For the global shared corpus, the primary raw layer achieved a CSSI of 0.9408, confirming a high level of baseline textual continuity among aligned units under the selected embedding model.

To evaluate the extreme margins of the vector field, the Tail Semantic Stability Index (***TSI***) tracks the remaining similarity at the upper bounds of the distance distribution, specifically isolating the 95th and 99th percentiles. The global corpus yielded a **$\mathbf{TSI_{95}}$** score of 0.8282 and a **$\mathbf{TSI_{99}}$** score of 0.7528. These metrics demonstrate high semantic persistence even within the most highly drifted segments of the historical text under the selected embedding model.

The major corpus divisions displayed highly parallel central stability behaviors. The Old Testament shared corpus division produced a median drift of 0.0589, resolving to a local CSSI of 0.9411. Concurrently, the New Testament shared corpus division yielded a median drift of 0.0601, resolving to a local CSSI of 0.9399. This symmetry reveals that semantic stability profiles remain highly uniform across highly distinct literary genres and compilation histories, with only minor fractional variance emerging between the major test blocks. Figure 2 maps the full ***R0*** distance distribution alongside the empirical percentile coordinates, and Table 2 summarizes the localized stability indices across each active corpus scope. A fuller work level view of median drift variation is provided in Appendix Figure A1, which reports the distribution of local intratextual stability across individual biblical works.

Table 2. Central and Tail Semantic Stability Indices by Corpus Scope

Scope	Aligned Units	Median Drift	CSSI	Tail / Interpretive Notes
Full shared corpus	31,102	0.0592	0.9408	P95 = 0.1718; P99 = 0.2472; TSI95 = 0.8282; TSI99 = 0.7528
Old Testament shared corpus	23,145	0.0589	0.9411	Intracorporal stability profile closely matches global corpus
New Testament shared corpus	7,957	0.0601	0.9399	Intracorporal stability profile closely matches global corpus

Note. Repository linked machine readable artifacts provide full precision values and row level outputs where applicable.

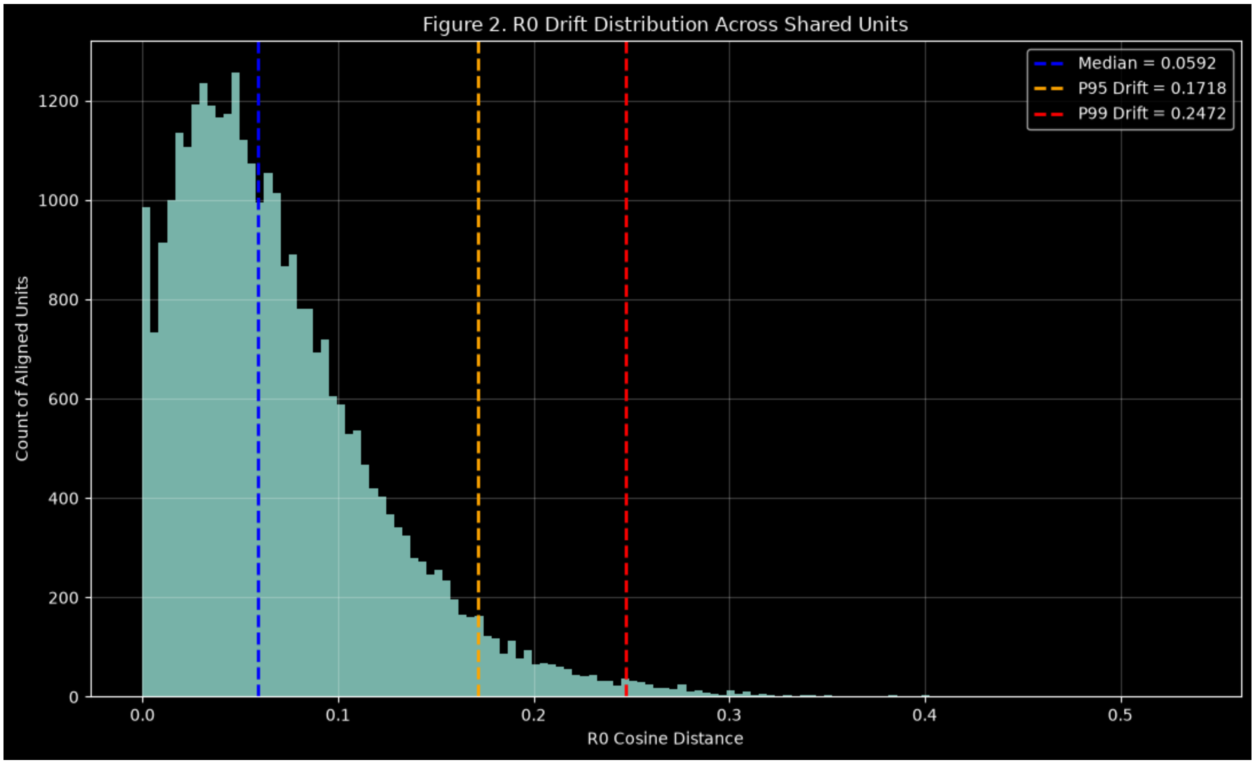


Figure 2. R0 drift distribution with median, 95th percentile, and 99th percentile markers

Embedding Invariance to Surface Normalization

The raw ***R0*** text layer and the preprocessed ***R1*** text layer generated identical pairwise distance scores for all 31,102 shared units. The maximum absolute difference between the two representation distance metrics was 0.0, and the global mean difference resolved to 0.0. This empirical equivalence reveals that the selected embedding model was invariant to the superficial normalization operations deployed in the ***R1*** workflow, specifically lowercasing and whitespace standardization.

Consequently, the two representation layers are not treated as independent statistical evidence streams. To prevent analytical redundancy and avoid artificially inflating the reported robustness of the system, the raw ***R0*** layer is selected as the exclusive reporting foundation for all downstream inferential modeling while the matching ***R1*** behavior is logged strictly as a baseline preprocessing invariance check. This distinction prevents the analysis from overstating empirical robustness across distinct representation conditions.

Hierarchical Thresholding and Drift Classification

By applying the local work level threshold boundaries established in the methodology, the pipeline isolated a tightly bounded upper tail population of units exhibiting higher distance values. Using the primary ***R0*** representative reporting layer, 29,505 aligned units were classified as stable inliers representing 94.87% of the global shared corpus. The secondary outlier boundary identified 1,238 units as review candidates, accounting for 3.98% of the data. The primary upper threshold classified 359 units as strong outliers, representing 1.15% of the total shared population. Together, review candidates and strong outliers accounted for 1,597 distinct units, or 5.13% of the shared canonical corpus.

Thresholds were assigned using the hierarchical precedence framework, which preferentially applies localized intratextual parameters derived directly from individual work properties before defaulting to broader corpus partitions. In this study, every shared unit met the density conditions required to resolve local intratextual parameters. Localization is statistically critical because it guarantees that the outlier classification of any individual verse is determined strictly relative to the unique variance profile of its parent work rather than being distorted by a single aggregate supracanonical threshold. The empirical distribution confirms that existence of an authentic but highly compressed drift tail, demonstrating that while the majority of the text remains safely within predictable bounds, a small specific subset requires manual human review. Table 3 provides a comprehensive summary of classification counts across each structural stratum. Figure 3 visualizes the proportional composition of the threshold classifications, showing that stable inliers constitute the majority of aligned units while review candidates and strong outliers form a compact upper-tail review population.

Table 3. Threshold Classification Counts for the Primary R0 Reporting Layer

Classification	Count	Share of Aligned Corpus	Interpretive Role
INLIER	29,505	94.87%	Stable aligned units below the secondary review threshold
REVIEW_CANDIDATE	1,238	3.98%	Units at or above the secondary threshold and below the primary outlier threshold
STRONG_OUTLIER	359	1.15%	Units at or above the primary outlier threshold
FLAGGED TOTAL	1,597	5.13%	Review candidates and strong outliers

Note. Repository linked machine readable artifacts provide full precision values and row-level outputs where applicable.

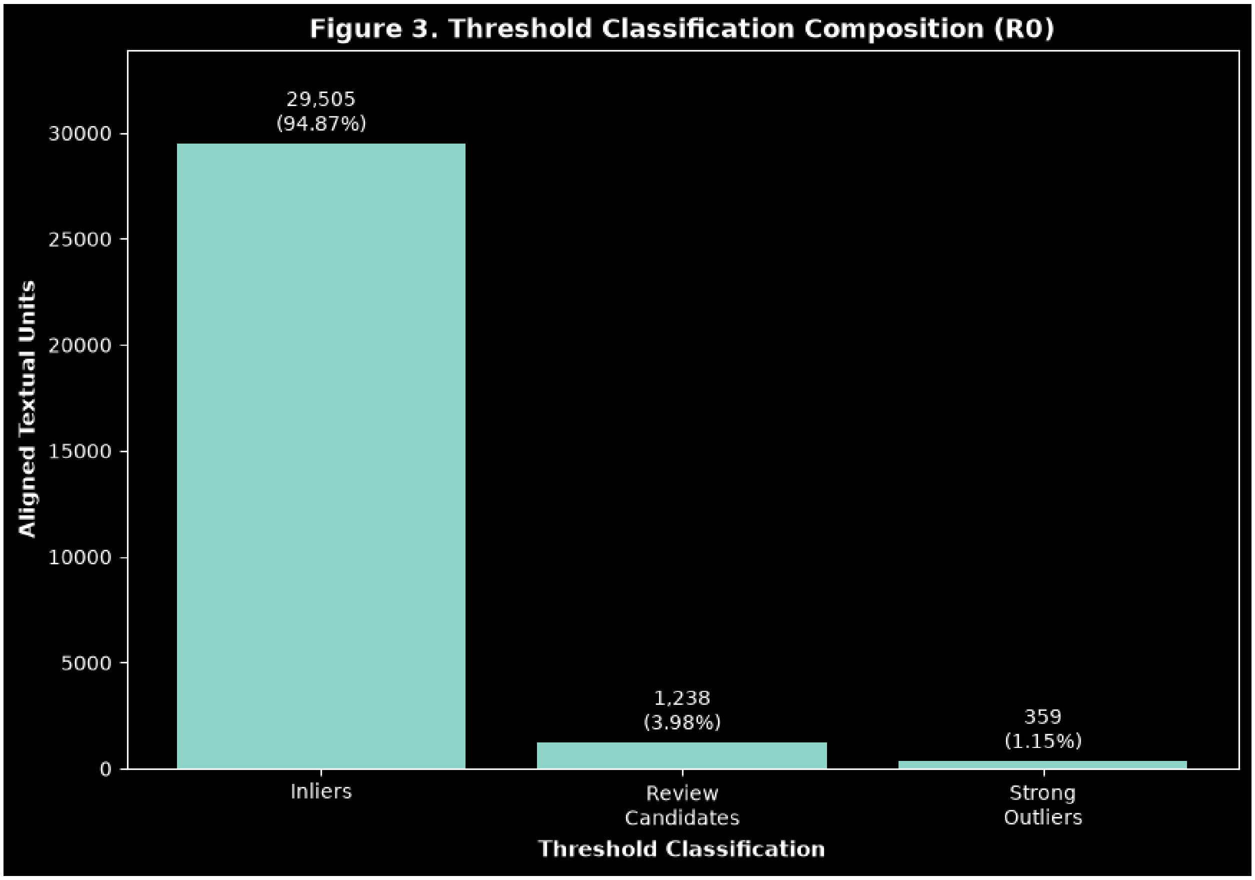


Figure 3. Proportional composition of robust threshold classifications in the primary R0 layer

Permutation Testing and the Stability Gap Baseline

The Monte Carlo permutation models reject the null hypothesis, confirming that the semantic proximity of true historically aligned pairs is substantially closer than randomized counterfactual arrangements. The primary intratextual null model executed 5,000 randomized iterations per work, while the secondary intracorporal and exploratory supracanonical configurations executed 1,000 and 500 iterations respectively to establish broad corpus baselines.

The absolute physical separation between historical continuity and the randomized background is quantified via the Stability Gap Index (***SGI***), which reflects the distance subtraction between the center of the empirical null distribution and the observed true median. Across all tested structural scopes and null variations, the authentic aligned medians fell drastically below the randomized centers, returning uniformly positive SGI values. To enable uniform scale comparisons across highly divergent text lengths and vocabulary sizes, the metric is standardized using the Normalized Stability Gap Index ($\boldsymbol{SGI^{norm}}$**),** which expresses the spatial separation as a direct proportion of the null model median.

The localized intratextual tests provide the definitive inferential foundation for the study. Every individual work level scope demonstrated observed distance scores below the randomized null distribution, yielding false discovery rate adjusted permutation p values below the critical 0.05 alpha threshold. The secondary intracorporal and global supracanonical validation steps displayed an identical directional behavior, confirming the statistical robustness of the text alignment across all scales of environmental randomization.

As formalized in the procedural layout, the calculation of the permutation p value incorporates an add one sampling correction to adjust for the absolute limits of finite Monte Carlo iterations. Under this adjustment, the 5,000 iteration intratextual pool sets a strict analytical floor of approximately 0.0002, while the 500 iteration supracanonical pool resolves to a baseline floor of approximately 0.0020. As a result, fractional differences in the lowest reported significance boundaries between the test layers represent mathematical iteration tracking constraints rather than variations in the underlying historical effect sizes.

The secondary distributional diagnostics were fully congruent with the primary permutation outcomes. Two sample Mann Whitney U rank calculations proved that the true aligned distances maintain significantly lower average ranks than their randomized null counterparts, while two sample Kolmogorov Smirnov tests revealed strong cumulative distribution separation between the authentic and counterfactual fields. Furthermore, Cliff's delta values were uniformly large and negative across all testing configurations, confirming that observed aligned distances were overwhelmingly lower than their randomized counterfactuals. Distribution wide indicators confirm comprehensive spatial divergence across the entire corpus, and they support the primary central tendency findings without introducing confounding sample size dependencies. True structural alignment is therefore demonstrated to be the primary driver of vector proximity ensuring that historical text pairings remain significantly closer than random alternatives across all structural dimensions. Table 4 outlines the permutation outcomes, and Figure 4 maps the observed distance medians against the null distribution fields.

Table 4. Monte Carlo Permutation Baseline Summary

Null model	Analytical role	Scope	Iterations	Minimum attainable p-value	Directional result
Intratextual	Primary	Work level	5,000	$1 / 5{,}001 \approx 0.000200$	Observed medians below null. All work level tests FDR adjusted $p < 0.05$
Intracorporal	Secondary	OT shared / NT shared	1,000	$1 / 1{,}001 \approx 0.000999$	Observed medians below null in the same directional pattern
Supracanonical	Exploratory	Full shared corpus	500	$1 / 501 \approx 0.001996$	Observed medians below null in the same directional pattern

Note. Repository linked machine readable artifacts provide full precision values and row-level outputs where applicable.

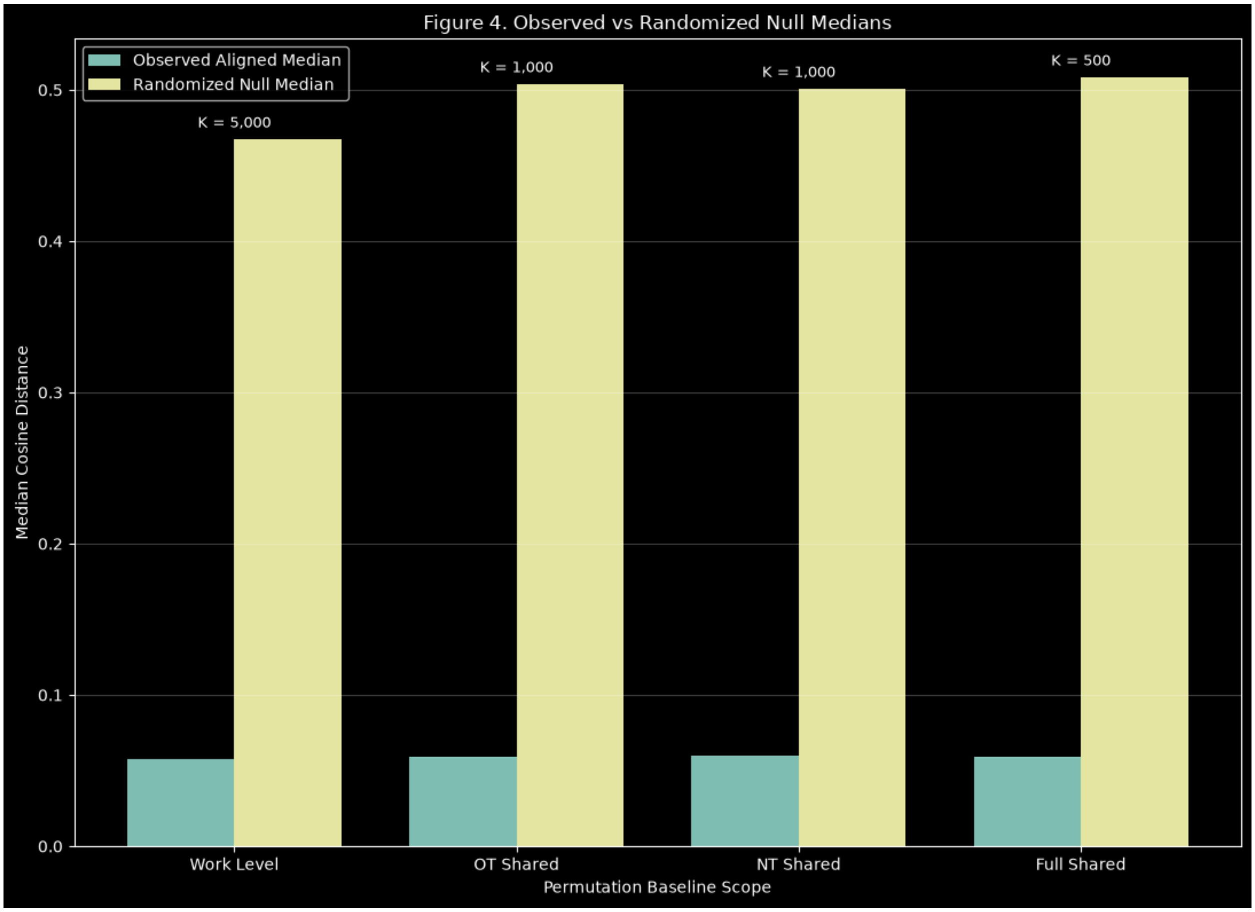


Figure 4. Observed aligned medians compared with randomized null median distributions

HITL Category Enrichment Analysis

Manual human in the loop annotations were utilized to interpret the exact linguistic properties driving the flagged tail units relative to the stratified background control sample. Due to the mathematical distance identity tracking between the active vector layers, the ***R0*** representation served as the sole representative for all primary category rate and enrichment reporting. While the full multi label observations were retained within a secondary unique unit union summary to preserve the qualitative records, individual ***R0*** and ***R1*** rows were isolated during statistical testing to avoid artificial sample size duplication. Table 5 outlines the underlying linguistic taxonomy schema.

Table 5. HITL Annotation Category Schema

Category	Operational Definition	Representative Example / Cue
orthographic_character_form	Spelling, historical character form, capitalization, abbreviation, ampersand expansion, or form level modernization that preserves lexical meaning and syntactic function.	Iesus → Jesus; vnto → unto; & → and
punctuation_clause_relation	A punctuation change that may alter clause relation, scope, coordination, explanation, contrast, continuation, or syntactic attachment without necessarily changing the lexical words.	: → ;, ; → ,, , → ., or other punctuation changes that affect whether a clause reads as explanation, continuation, coordination, apposition, or a new syntactic unit.

lexical_substitution	One word or phrase is replaced with a different lexical item that could affect nuance, specificity, reference, or meaning.	A changed noun, verb, adjective, title, theological term, or phrase where the replacement is not merely spelling modernization.
function_word_change	Changes to articles, prepositions, conjunctions, particles, or other grammatical function markers.	in → into, of → from, and → but, added/removed the, a, to, for, or similar grammatical-word changes.
pronoun_shift	A change involving pronouns, possessive pronouns, antecedent clarity, person, number, gender reference, or pronominal ambiguity.	he → they, his → their, thou/thee/ye/you changes, or a pronoun becoming more or less ambiguous.
negation_modality_change	A change involving negation, obligation, permission, possibility, certainty, command, tense/aspect force, or modal meaning.	shall, will, may, might, must, cannot, not, never, no, or changes affecting imperative, conditional, prohibitive, or certainty force.
word_order_syntactic_shift	Alterations in the sequence or syntactic ordering of constituents.	The same basic words are present, but their order, grouping, or syntactic attachment changes.
addition_deletion	Material is added, removed, or substantially expanded/contracted compared with the paired edition.	A word, phrase, clause, name, title, explanatory element, or repeated phrase appears on one side but not the other.
multi_factor_complex	Overlapping or compounded changes where multiple categories interact within a single unit.	Mixed lexical, syntactic, and punctuation changes

Note. Repository linked machine readable artifacts provide full precision values and row level outputs where applicable.

The categorical analysis revealed that the high distance drift tail is overwhelmingly governed by structural orthographic and character form modernization. The specific category labeled *orthographic_character_form* was present across the flagged outlier population and was the only linguistic taxonomy marker to achieve significant statistical overrepresentation after strict false discovery rate correction. This category encompasses mechanical adjustments such as spelling modernization, letterform changes, capitalization changes, abbreviation expansion, ampersand expansion, and similar form level changes that do not alter lexical meaning or syntactic function.

The remaining taxonomic categories were descriptively present but failed to show statistically significant enrichment within the flagged outlier group when evaluated against the background control sample. The *punctuation_clause_relation* marker appeared within a meaningful minority of flagged units, reflecting punctuation changes that may alter clause relation, scope, coordination, explanation, contrast, continuation, or syntactic attachment. However, its occurrence rate within the drift tail was statistically indistinguishable from its baseline occurrence rate within the stable background sample, failing to support a valid overrepresentation claim. A matching behavior was recorded for the *lexical_substitution* category, which appeared in both the outlier tail and the background sample without reaching significance after multiple comparison adjustments.

These findings partially support the category structure hypothesis. The drift tail does exhibit category structure, but the dominant structure is orthographic and character form modernization rather than broad lexical, syntactic, or doctrinal semantic divergence. Table 6 summarizes the frequentist overrepresentation tests and odds ratios, and Figure 5 compares category rates between flagged units and the inlier background sample.

Table 6. HITL Category Enrichment and Overrepresentation Summary Using Unique Unit Counts

Category	Flagged count	Flagged rate	Background count	Background rate	FDR Significant Overrepresentation
orthographic_character_form	1,597 / 1,597	100.0%	519 / 526	98.7%	Yes

punctuation_clause_relation	277 / 1,597	17.3%	82 / 526	15.6%	No
lexical_substitution	197 / 1,597	12.3%	83 / 526	15.8%	No
addition_deletion	11 / 1,597	0.7%	1 / 526	0.2%	No
function_word_change	2 / 1,597	0.1%	1 / 526	0.2%	No
pronoun_shift	1 / 1,597	0.1%	0 / 526	0.0%	No
word_order_syntactic_shift	1 / 1,597	0.1%	0 / 526	0.0%	No
negation_modality_change	0 / 1,597	0.0%	0 / 526	0.0%	No
multi_factor_complex	0 / 1,597	0.0%	0 / 526	0.0%	No

Note. Repository linked machine readable artifacts provide full precision values and row level outputs where applicable.

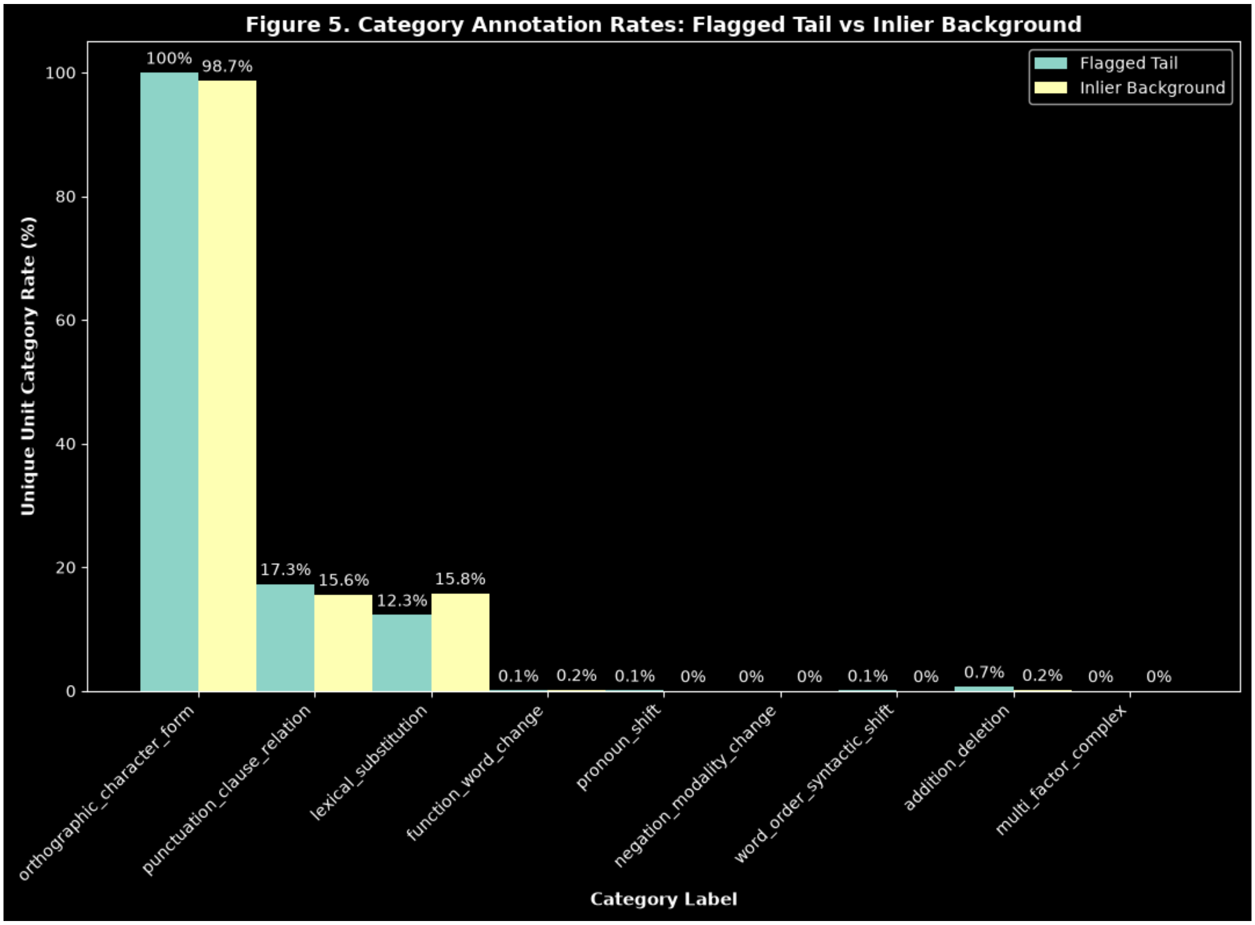


Figure 5. HITL category rates for flagged units and the inlier background sample

Evaluation of Representative High Drift Textual Examples

The representative text records illustrate how the numeric outlier rankings translate to human readable textual transformations. The most extreme outliers isolated by the pipeline include specific historical alignments such as Mark 14:17, 1 Thessalonians 5:16, Leviticus 11:15, Matthew 27:14, Deuteronomy 14:14, 2 Timothy 3:7, Luke 19:1, John 2:13, Lamentations 5:8, and 1 Chronicles 23:27.

Many of these high drift cases are driven by early modern orthographic conventions in the KJV 1611 text that underwent systematic modernization in the KJV 1769 edition. Prominent repetitive shifts include the stabilization of typography such as “Iesus” to “Jesus”, “euening” to "evening", "hee" to "he", "vnto" to "unto", and "Rauen" to "raven". A smaller portion of these archetypes incorporate distinct clause relation adjustments visible in colon to semicolon transitions or punctuation modernizations that reshape how independent propositions are segmented and visually associated.

These qualitative examples demonstrate that elevated embedding distances should not be casually misread as indicators of theological, conceptual, or doctrinal semantic divergence. Instead, the highest distances often capture the technical interaction between modern language model architectures and early modern English orthographic, typographic, and character level variations. Table 7 lists the ten representative high drift examples alongside their canonical references, raw scalar distance scores, local threshold assignments, human verified change categories, and observed text level changes.

Table 7. Representative High Drift Textual Examples

Unit Reference	R0 Drift	Classification	Primary Observed Change Pattern	Observed Text Level Changes
MRK:14:17	0.5351	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace**: [euening hee commeth] -> [evening he cometh] **Replace**: [twelue.] -> [twelve.]
1TH:5:16	0.5124	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace**: [Reioyce euermore:] -> [Rejoice evermore.]
LEV:11:15	0.5035	STRONG_OUTLIER	Orthographic / Character form modernization, Punctuation	**Replace**: [Euery Rauen] -> [Every raven] **Replace:** [kinde:] -> [kind;]
MAT:27:14	0.4950	STRONG_OUTLIER	Orthographic / Character form modernization, Punctuation	**Replace**: [neuer] -> [never] **Replace**: [word:] -> [word;] **Replace**: [Gouernour marueiled] -> [governor marvelled]
DEU:14:14	0.4918	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace**: [euery rauen] -> [every raven] **Replace**: [kinde,] -> [kind,]
2TI:3:7	0.4668	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace:** [Euer] -> [Ever] **Replace**: [neuer] -> [never] **Replace:** [trueth.] -> [truth.]
LUK:19:1	0.4653	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace:** [Iesus entred,] -> [Jesus entered] **Replace**: [thorow Iericho.] -> [through Jericho.]
JHN:2:13	0.4308	STRONG_OUTLIER	Orthographic / Character form modernization	**Delete:** [¶] -> [] **Replace**: [Iewes Passeouer] -> [Jews' passover] **Replace:** [& Iesus] -> [and Jesus] **Replace:** [vp] -> [up] **Replace:** [Hierusalem] -> [Jerusalem,]
LAM:5:8	0.4197	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace:** [Seruants haue] -> [Servants have] **Replace**: [ouer vs:] -> [over us:] **Replace**: [doeth deliuer vs] -> [doth deliver us]
1CH:23:27	0.4120	STRONG_OUTLIER	Orthographic / Character form modernization	**Replace**: [Dauid,] -> [David] **Replace**: [Leuites] -> [Levites] **Replace**: [numbred] -> [numbered] **Replace:** [twentie yeeres olde,] -> [twenty years old] **Replace:** [aboue:] -> [above:]

Note. Repository linked machine readable artifacts provide full precision values and row level outputs where applicable.

Synthesis of Semantic Stability Findings

The combined empirical findings confirm that the KJV 1611 and KJV 1769 editions are highly stable semantically at the individual aligned unit level under the selected embedding model. The shared corpus achieved total structural alignment alongside total vector coverage across the active representation profiles. Median drift values remained consistently low across all scopes, the CSSI remained high, and Monte Carlo permutation testing proved that true authentic historical text alignments remain substantially closer than randomized counterfactual pairings across intratextual, intracorporal, and supracanonical baselines.

Concurrently, robust hierarchical thresholds successfully isolated a compact and highly specific upper tail consisting of review candidates and strong outliers. Human in the loop category enrichment and overrepresentation analysis revealed that this drift tail is dominated almost exclusively by formal orthographic and character level modernization. These insights validate the InterLink workflow as a robust methodology for automated text alignment and linguistic comparison while highlighting clear avenues for future research, including the deployment of aggressive historical orthographic normalization engines, KJV 1611 exclusive corpus analysis, and the training of domain specific theological embedding architectures.

Discussion

Interpretation of the Core Findings

This study demonstrates that the King James Version (KJV) 1611 and KJV 1769 editions maintain high central semantic stability at the individual aligned textual unit level under the selected embedding model. The comparison corpus achieved complete structural alignment across 66 shared canonical works establishing a firm computational foundation across 31,102 unique unit pairings with total vector coverage across the active representation layers. This design allows the framework to distinguish three closely related but independent empirical outcomes: the editions are centrally stable, a small upper tail of elevated drift units exists, and the primary drivers of that tail are dominated by formal orthographic and character form modernization rather than doctrinal semantic divergence.

The central stability finding is supported by the low median drift value and the resulting high Central Semantic Stability Index (CSSI). Across the full shared corpus, the median embedding distance remained remarkably low, while the Old Testament and New Testament divisions produced highly similar stability profiles. This distributional symmetry indicates that semantic continuity is not restricted to a single major corpus division, literary genre, or localized textual environment. Instead, the observed semantic stability appears broadly distributed across the shared KJV 1611 and KJV 1769 corpus.

Concurrently, the analytical framework avoids collapsing the entire corpus into a single generalized claim of text uniformity. The robust intratextual threshold boundaries successfully identified a bounded upper tail consisting of localized review candidates and strong outliers. While this tail represents a small statistical minority of the global shared corpus, it is methodologically important because it ensures the analytical pipeline retains the granular sensitivity required to inspect localized contextual deviations rather than allowing a high aggregate stability metric to obscure meaningful variation. The combination of high central stability and a measurable drift tail constitutes one of the core insights of this study.

Significance of the Permutation Baseline Results

The Monte Carlo permutation models provide definitive inferential evidence for the geometric validity of the aligned comparison. Across the intratextual, intracorporal, and supracanonical null models, true historically aligned text pairs were substantially closer in vector space than any randomized counterfactual pairings. This global divergence demonstrates that the measured proximity between the KJV 1611 and KJV 1769 units is not merely an artifact of comparing two texts drawn from the same religious corpus or uniform biblical vocabulary pool. Rather, the historical structural alignment itself carries the measurable semantic significance.

The intratextual null model provides the most conservative and localized inferential baseline for the study because the randomized target selection pool was restricted strictly to the boundaries of the same biblical work, the permutation routine preserved local vocabulary densities, book specific contextual profiles, and individual literary styles. Even when subjected to this rigorous constraint the true aligned units consistently demonstrated significantly lower distances than the randomized alternatives. The Stability Gap Index (***SGI***) and normalized SGI further quantify this separation, showing that authentic historical alignment substantially reduces semantic distance relative to randomized pairings.

The secondary distributional diagnostics reinforce these permutation outcomes. The Mann Whitney rank sum tests, Kolmogorov Smirnov cumulative distribution comparisons, and Cliff's delta metrics all showed that the observed aligned distance distributions are not slightly shifted at their center. Instead, they are comprehensively separated from the empirical null fields across their entire range and dispersion profiles. Although these diagnostics are treated as secondary checks rather than the primary inferential benchmarks, their mathematical alignment with the permutation results strengthens the conclusion that structural alignment is the definitive driver of vector proximity.

Interpretation of the Drift Tail

The thresholded drift tail should be interpreted as a review population rather than as immediate evidence of conceptual, theological, or doctrinal divergence. The hierarchical pipeline identified 1,238 review candidates and 359 strong outliers under the primary representation layer, together representing 5.13% of the shared corpus. These classifications indicate that the affected textual units exceeded robust localized drift expectations, not that they necessarily altered theological meaning.

This analytical distinction is critical for downstream interpretation. Embedding distance operates strictly as a geometric measurement calculated within a model defined vector space. While it serves as an excellent diagnostic engine for isolating units that deviate from expected variance fields, it is not inherently self-interpreting. An elevated pairwise distance score may reflect a genuine lexical substitution or syntactic change, but it may also reflect superficial variations in orthographic modernization, punctuation structure, typography, capitalization, abbreviation expansions, or early modern spelling conventions. For this reason, the methodology treats thresholded outliers as candidates for qualitative human inspection rather than as finalized semantic evaluations.

The implementation of localized intratextual parameters improves the interpretability of this flagged tail. Given that all unit level classifications were determined relative to local work level thresholds, each shared verse was evaluated exclusively against the unique variance of its own biblical work. This structural localization prevents a single global threshold from over flagging stable units in books with naturally higher drift distributions or under flagging meaningful mutations in books with naturally lower drift distributions. This results in a drift tail that functions as a highly contextualized and locally sensitive review set.

Orthographic and Character Form Modernization Significance

The human in the loop (HITL) category enrichment analysis clarifies the nature of the drift tail. The overriding structural pattern emerged around formal orthographic and character form modernization. The taxonomy category labeled *orthographic_character_form* was the sole linguistic marker to achieve statistically significant overrepresentation after false discovery rate adjustment. This result demonstrates that the highest distance units were primarily characterized by spelling modernizations, letterform adjustments, capitalization changes, and ampersand or abbreviation expansions instead of driven by lexical replacement, doctrinal semantic alteration, or widespread syntactic restructuring.

This finding carries major methodological implications for researchers interpreting embedding-based drift across historical or sacred texts. Modern embedding models are trained primarily on modern or normalized text distributions [5]. As a result, authentic early modern spellings and archaic character forms can produce measurable distance even when the underlying lexical identity or theological meaning remains unchanged. Examples such as “Iesus” to “Jesus”, “vnto” to “unto”, “euening” to “evening”, and “hee” to “he” illustrate how surface level typography can influence vector behavior without indicating a true semantic divergence.

The descriptive visibility of the *punctuation_clause_relation* and *lexical_substitution* categories remains useful for mapping the full landscape of the corpus. However, because these categories failed to achieve statistically significant overrepresentation relative to the stratified background control sample, they cannot be leveraged to explain the emergence of the drift tail. Their presence indicates that the corpus contains complex variations beyond simple spelling changes, but the enrichment models prove that these deeper structural markers do not uniquely differentiate the outlier tail from the stable body of the text. The underlying research hypotheses are therefore clarified and partially supported: the drift tail exhibits definite categorical structure, but that structure is heavily concentrated around formal orthographic variations rather than conceptual or theological changes.

Embedding Invariance to Light Normalization

The identical distance scores recorded across the raw and preprocessed representation layers constitute a separate finding. The application of light normalization (specifically restricted to lowercasing and whitespace standardization) produced identical pairwise drift metrics across all 31,102 shared units. This indicates that the selected embedding model was invariant to these light preprocessing operations within this computational workflow.

This outcome should not be misconstrued as evidence that text normalization is generally irrelevant to vector processing. The outcome establishes that the specific normalization operations used in ***R1*** were not sufficient to alter the embedding behavior of the model for this edition pair. The overwhelming dominance of orthographic and character form variation revealed by the manual annotations suggests that more aggressive, historically aware normalization routines are required for future analytical iterations. Such preprocessing layers must directly target early modern spelling patterns, archaic letterform conventions, typographic variations, and systematic punctuation regularizations.

The empirical equivalence across the representations justifies the reporting policy used in the study because the raw and lightly normalized layers generated matching distance vectors. Treating them as separate streams of statistical evidence would violate modeling integrity. The raw representation was isolated as the sole layer for inferential reporting, while the secondary layer was maintained strictly as an invariance verification check. This design choice prevents artificial sample size inflation and ensures the study does not overstate its environmental diversity across representation conditions.

Implications for Computational Theological Text Analysis

These cumulative findings validate the InterLink pipeline as a practical end to end architecture for computational theological text comparison. The workflow proves that the platform can ingest structured historical corpora, perform structural alignment, generate representation specific embeddings, isolate local outliers via robust thresholds, compute empirical permutation baselines, generate drift metrics, and integrate human annotations for interpretive grounding. This integrated architecture is significant because it connects large scale quantitative distance measurement with human readable qualitative evaluation.

The study also provides an essential cautionary warning against relying on embedding similarity metrics in isolation. Embedding distance functions as an exceptionally efficient discovery engine for prioritizing anomalous text blocks, but it cannot determine on its own whether a specific vector displacement stems from an orthographic, syntactic, lexical, or doctrinal modification [6]. The manual annotation layer is therefore essential for separating model sensitivity from textual significance.

For the specific comparison between the KJV 1611 and KJV 1769 editions, the workflow found no evidence of broad doctrinal semantic divergence at the aligned unit level. However, this is best stated carefully. The study does not prove that every interpretive issue between the editions is exhausted by embedding distance, nor does it replace traditional philology or textual criticism. Instead, the study shows that under the selected model, representation layers, alignment rules, and annotation schema, the shared corpus is centrally stable, and the drift tail is largely explained by formal modernization.

Implications for Historical Orthographic Normalization

The dominance of orthographic and character form variation motivates a dedicated normalization follow up study. The lightly preprocessed representation layer was deliberately minimal, omitting deep historical transformations like mapping early modern spelling rules into modern equivalents, regularizing u/v or i/j substitutions, expanding archaic typography in a historically aware manner, or correcting for systematic early modern printing conventions.

A future aggressive normalization layer should test whether the measured drift tail contracts when historical forms are mapped to modern typographical equivalents. If the volume of strong outliers and review candidates decrease substantially under such normalization, that contraction would support the interpretation that much of the current tail reflects model sensitivity to historical letterforms rather than semantic variance. Conversely, if a compact residual tail persists after comprehensive normalization, those remaining units would become especially important targets for philological and theological review.

This future work would also help separate two different kinds of stability: surface form stability and semantic content stability. The present study shows high semantic stability even while preserving raw historical variations. A fully normalized future analysis will complement this by determining how much of the remaining vector separation is attributable to orthographic style rather than deeper lexical or syntactic difference.

Treatment of KJV 1611 Exclusive Material

The KJV 1611 exclusive corpus was inventoried but excluded from the paired drift analysis because it lacks corresponding KJV 1769 shared canon counterparts. This decision preserves the integrity of the comparative design. Direct drift analysis requires paired textual elements that occupy the same structural location across both historical canons. Since the exclusive material does not satisfy this structural requirement, its inclusion in the primary drift equations would conflate alignment-based comparison with generalized cross corpus similarity matching.

Nevertheless, the exclusive corpus remains important for the broader InterLink research agenda. Its inventory establishes the basis for future nearest neighbor analysis, in which the KJV 1611 exclusive works can be positioned relative to shared canon regions using semantic retrieval instead of direct structural alignment. This future analysis would address the H4 research hypotheses by identifying where these unaligned historical works sit semantically within the canonical landscape and determining whether their vector positions cluster near specific literary genres, topical domains, or theological frameworks.

Limitations

Several core limitations must guide the application of these conclusions.

Model Architecture Constraints

The study relies on a single embedding model, mxbai-embed-large. The empirical metrics directly reflect the specific behavior of this individual model under the chosen representation layers and should not be assumed to generalize automatically to all alternative embedding architectures. Different model families may display divergent sensitivities when encountering early modern English typography, punctuation boundaries, capitalization styles, or localized biblical language [7].

Normalization Scope

The comparative framework only evaluated light normalization. Since the ***R1*** layer did not incorporate historical spelling transformations or advanced orthographic normalization, this study cannot empirically determine how much of the drift tail would contract under more aggressive normalization. The finding that the raw and lightly cleaned rows yielded identical distances is a valuable technical finding, but it cannot serve as a substitute for a true historical normalization experiment.

Computational Stratification of the Permutation Architecture

The Monte Carlo permutation architecture introduces a computational constraint. To maintain uniform probability resolution across all testing strata every null model configuration would ideally be evaluated using the same high iteration count, such as 5,000 iterations per scope to provide uniform p-value resolution across intratextual, intracorporal, and supracanonical tests. However, the computational cost of randomized re-pairing increases exponentially as the valid candidate selection pool expands across larger structural boundaries. To preserve processing efficiency, iteration volumes were stratified by inferential priority: the primary intratextual baseline utilized 5,000 iterations, the secondary intracorporal check used 1,000 iterations, and the exploratory supracanonical validation used 500 iterations. This design preserves high statistical resolution for the critical work level hypothesis tests while providing broad corpus checks at reduced computational cost. As a result, p-value floors differ by null model level, and direct comparison of minimum p-values across null models should be avoided. Differences in the smallest attainable p-values partly reflect iteration count constraints rather than differences in underlying effect magnitude.

Annotation Schema Boundaries

The HITL annotation framework depends on the selected category schema. The schema was designed to capture major visible change types, but any annotation system imposes interpretive boundaries. Rare categories such as pronoun shifts, negation modality adjustments, constituent word order re-orderings, additions or deletions, or highly complex multi factor transformations may require expanded annotation coverage in future studies.

Control Sample Design

The background control sample was structured as a deterministic stratified sample rather than an exhaustive annotation census of all stable inliers. This design kept manual evaluation tracking tractable while supporting enrichment and overrepresentation analysis, but it does not supply a complete census of every change category across the stable body of the corpus.

Granularity and Context Window Limits

The study operates at the individual aligned unit level and does not directly model larger discourse structures, paragraph level semantics, cross references, theological themes, or doctrinal systems. A verse level or unit level comparison is appropriate for this method validation study, but future research should explore whether broader context windows produce different interpretive patterns.

Future Work

Historical Orthographic Normalization

Introducing a representation layer that maps early modern spellings, letterforms, ampersands, and typographic conventions into normalized forms before embedding, and then directly comparing the R0 baseline against this representation will reveal how much of the measured drift tail stems strictly from historical orthography.

Semantic Retrieval and Canonical Mapping

Nearest neighbor analysis of the exclusive KJV 1611 corpus will address texts that lack direct aligned pairs. Utilizing semantic retrieval against the shared canon will allow the InterLink platform to identify semantically proximate regions and evaluate whether these relationships are structured by genre, theme, theology, or lexical field.

Domain Specific Embedding Models

Developing domain specific theological embedding models will improve semantic precision. While general purpose embeddings identify broad semantic stability, they remain sensitive to historical orthography. Models trained directly on biblical, early modern, theological, and historical corpora will better distinguish surface variation from conceptual or doctrinal content.

Cross Corpus Expansion

Scaling the architecture to cross corpus theological research represents the final trajectory. While the present study validates the aligned edition workflow on a tightly controlled KJV 1611 and KJV 1769 comparison, the InterLink framework will extend to heterogeneous corpora, including the Septuagint, Dead Sea Scrolls, Patristics, and apocryphal literature. This phase will leverage the platform's existing provenance and metadata tracking to manage non-uniform textual structures, multiple languages, and uncertain alignments while preserving human interpretive review.

Conclusion

The overall findings support a balanced, scientific interpretation of the historical text layers. The KJV 1611 and KJV 1769 editions display exceptional semantic stability at the aligned unit level under the selected embedding model. True historical aligned pairs are substantially closer than randomized alternatives, and hierarchical threshold models identify only a compact upper tail of review candidates and strong outliers. Human annotation shows that this tail is dominated by orthographic and character form modernization rather than systemic doctrinal or semantic divergence.

The study therefore validates both the stability of the edition pair, and the usefulness of the InterLink computational workflow. At the same time, it highlights an important methodological caution that modern embedding distances can reflect historical surface form variation, and high drift values should always be interpreted through structured human review. This integration of computational measurement and human interpretation provides a foundation for future computational theological research.

Data and Code Availability

The study artifacts, generated outputs, diagnostic files, annotation datasets, architecture documentation, validation notebook, exploratory analysis materials, and figure generation materials are available in the public versioned research repository associated with this paper: https://github.com/Saorsa-Ind/kjv-1611-1769-semantic-stability-study.

Large machine-readable artifacts are not reproduced in full within the manuscript appendices due to their size. Instead, the appendices provide artifact descriptions and repository relative paths. The repository manifest provides the complete artifact map and identifies the source files used to support the manuscript tables and figures.

The proprietary InterLink Research Platform source code is not publicly released. The released notebooks and generated artifacts are provided to support independent inspection of the reported analyses, reproduction of the manuscript figures, verification of the manuscript tables, and review of the empirical outputs used in this study. Upon archival release, the repository should be cited using the GitHub repository and release tag identified in Appendix G. The associated archival DOI is provided by the Zenodo record for that release.

References

[1] Limburg, David, et al., editors. *The Oxford Handbook of the Bible and Textual Criticism*. Oxford UP, 2023.

[2] Manjavacas, Enrique, et al. "Automatic Detection of Complex Quotation Patterns in Aggadic Literature." *Cogent Arts & Humanities*, vol. 13, no. 1, 2026, pp. 1–18.

[3] Enevoldsen, Kenneth, et al. "The Scandinavian Embedding Benchmarks: Comprehensive Assessment of Multilingual and Monolingual Text Embedding." *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 40336–40358. *NeurIPS Proceedings*, neurips.cc/virtual/2024/poster/97869.

[4] Cocheril, Jean-Baptiste, et al. "How Surprising Is Historical Italian to Language Models? A Diagnostic Framework for Linguistic Drift." *arXiv preprint arXiv:2606.27275*, 2026, pp. 1–15.

[5] Nayeem, Mir Tafseer, and Davood Rafiei. "Which English Do LLMs Prefer? Triangulating Structural Bias Towards American English in Foundation Models." *arXiv*, 2026, arXiv:2604.04204.

[6] Kovács, Gábor, and J. Alonso. "Anomaly Detection in Texts Using Sentence Embeddings." *Proceedings of the Workshop on Natural Language Processing for Outlier Detection*, edited by Workshop Chairs, CEUR-WS, 2023, pp. 1-12. *CEUR Workshop Proceedings*, vol. 3498, paper 4, ceur-ws.org/Vol-3498/paper4.pdf.

[7] Nguyen, Thang, et al. "Tokenization is Sensitive to Language Variation." *arXiv preprint arXiv:2502.15343*, 2025, pp. 1-15.

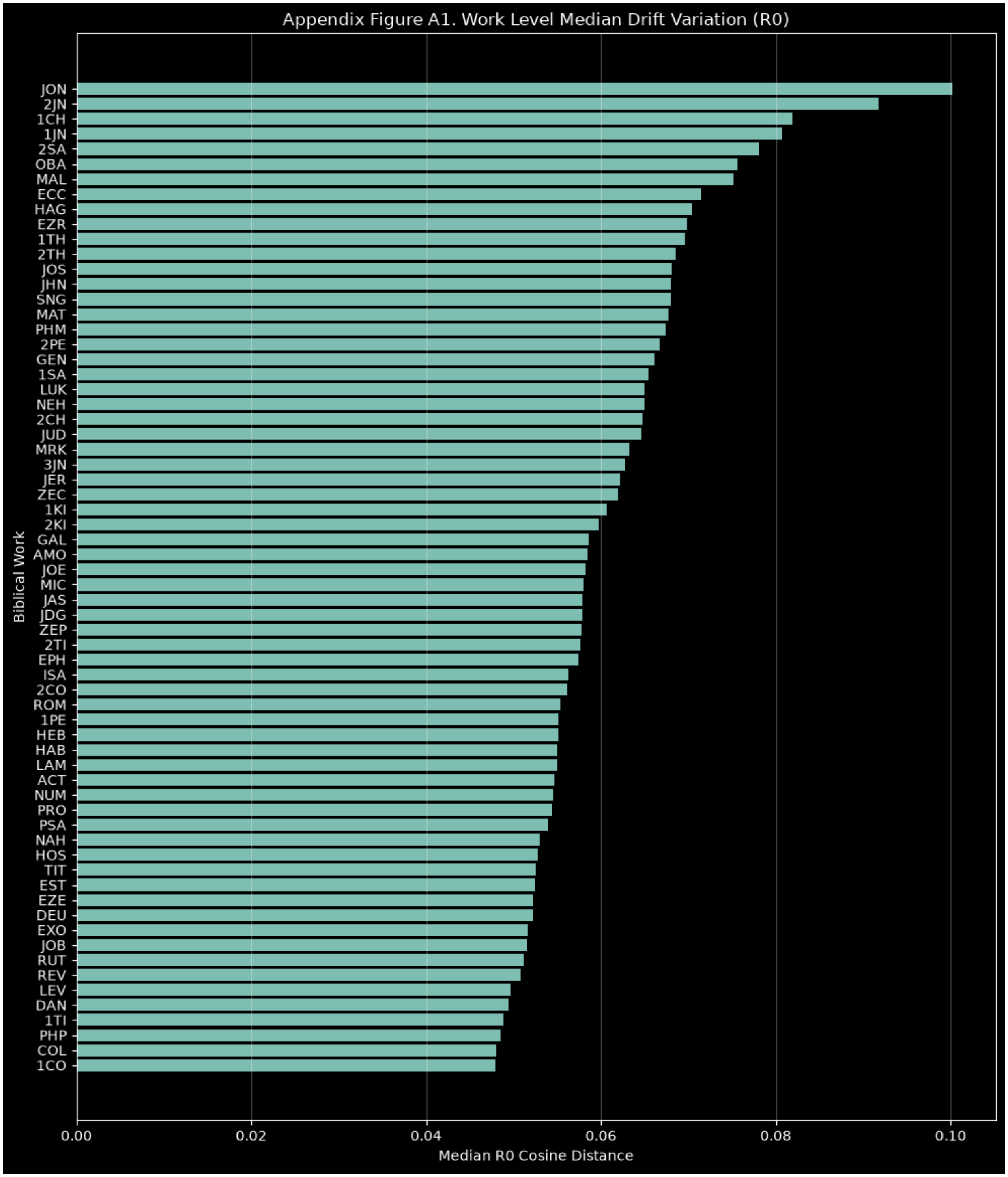
Appendix A: Full Intratextual Drift Index Summary

This appendix indexes the drift summary artifacts used to support the study's central and tail semantic stability findings. The listed files provide scope level drift indices for the shared corpus, including work level, work group level, and full shared canon summaries. These artifacts preserve the full precision values underlying the manuscript's stability metrics and provide the source data for the work level variation visualization reported in Appendix Figure A1.

Field	Description/Value
Artifact	drift_indices_by_scope.csv, sensitivity_indices_by_scope.csv, scope_registry.csv, scope_registry.parquet, appendix_figure_A1_work_level_drift_variation.png
Repository Path(s)	data/drift/drift_indices_by_scope.csv, data/drift/sensitivity_indices_by_scope.csv, data/coverage/scope_registry.csv, data/coverage/scope_registry.parquet, figures/appendix_figure_A1_work_level_drift_variation.png
Purpose	Provides intratextual, intracorporal, and supracanonical drift indices used to summarize central and tail semantic stability, representation sensitivity, and work level drift variation.
Manuscript use	Supports Table 2 and Appendix Figure A1.

Appendix Figure A1: Work Level Drift Variation

Appendix Figure A1 visualizes work level median drift variation across the shared biblical works. The figure supplements the corpus level stability indices by showing how local intratextual drift varies by individual work while remaining within the broader semantic stability profile reported in the main results.



Appendix Figure A1. Work Level Median Drift Variation Across Biblical Works

Appendix B: Threshold Summary by Scope and Representation

This appendix indexes the thresholding artifacts generated by the robust stratified classification framework. The listed files preserve the primary and secondary threshold boundaries, threshold source metadata, safeguard indicators, and unit level drift labels used to classify aligned units as inliers, review candidates, or strong outliers. These artifacts provide the machine-readable basis for the threshold classification counts and upper tail review population discussed in the Results section.

Field	**Description/Value**
Artifact	threshold_summary_by_scope.csv, drift_labels_by_unit.csv
Repository Path(s)	data/thresholds/threshold_summary_by_scope.csv, data/thresholds/drift_labels_by_unit.csv
Purpose	Records primary and secondary threshold boundaries, percentile components, MAD components, safeguard flags, threshold source metadata, and unit level classifications.
Manuscript use	Supports Table 3 and the thresholding results.

Appendix C: Complete Outlier Rankings

This appendix indexes the ranked outlier artifacts used for qualitative review and representative example selection. The listed files contain the review candidate and strong outlier records sorted by drift magnitude, along with supporting interpretability fields and annotated high drift examples. These artifacts provide the traceable source records behind the representative high drift examples reported in Table 7.

Field	**Description/Value**
Artifact	outlier_rankings.csv; top_outliers_annotated.csv; interpretability_dataset.csv
Repository Path(s)	data/thresholds/outlier_rankings.csv, data/thresholds/top_outliers_annotated.csv, data/hitl/interpretability_dataset.csv
Purpose	Provides ranked review-candidate and strong-outlier records, annotated high-drift examples, diff summaries, and interpretability fields used for representative example selection.
Manuscript use	Supports Table 7 and representative example selection.

Appendix D: Permutation Baseline Outputs and Statistical Diagnostics

This appendix indexes the permutation baseline and statistical diagnostic artifacts used to evaluate whether true aligned KJV 1611 and KJV 1769 units are closer in embedding space than randomized alternatives. The listed files preserve observed vs null stability gap metrics, null distribution summaries, permutation test results, and secondary non-parametric diagnostics. These artifacts provide machine-readable evidence supporting the permutation results summarized in Table 4 and visualized in Figure 4.

Field	**Description/Value**
Artifacts	baseline_gap_indices_by_scope.csv, null_distribution_summary.csv, statistical_test_results.csv, statistical_diagnostics_by_scope.csv
Repository Path(s)	data/permutation/baseline_gap_indices_by_scope.csv, data/permutation/null_distribution_summary.csv, data/permutation/statistical_test_results.csv, data/permutation/statistical_diagnostics_by_scope.csv
Purpose	Provides observed vs null stability gap metrics, null distribution summaries, permutation test results, and secondary Mann-Whitney, Kolmogorov-Smirnov, and Cliff’s delta diagnostics.
Manuscript use	Supports Table 4 and Figure 4

Appendix E: HITL Annotation Outputs and Unique Unit Inventory

This appendix indexes the human-in-the-loop annotation artifacts used to interpret the linguistic change patterns associated with the drift tail. The listed files contain the full annotation dataset, stratified inlier background sample, category count summaries, and enrichment test outputs. These artifacts provide the evidence base for the HITL category schema, overrepresentation results, and category rate visualization reported in the main manuscript.

Field	**Description/Value**
Artifacts	category_annotation_dataset.csv, inlier_background_annotation.csv, category_counts.csv, category_enrichment_results.csv
Repository Path(s)	data/hitl/category_annotation_dataset.csv, data/hitl/inlier_background_annotation.csv, data/hitl/category_counts.csv, data/hitl/category_enrichment_results.csv
Purpose	Contains the full HITL annotation dataset, stratified inlier background annotation sample, category count summaries, and category overrepresentation test outputs.
Manuscript use	Supports Tables 5–6 and Figure 5.

Appendix F: KJV 1611 Exclusive Corpus Inventory

This appendix indexes the KJV 1611 exclusive corpus inventory. These records were excluded from direct aligned-pair drift calculations. The inventory preserves the exclusive material as a separate corpus partition for future nearest-neighbor, thematic clustering, and cross corpus semantic positioning analysis.

Field	Description/Value
Artifact	exclusive_dataset_inventory.csv
Repository Path(s)	data/exclusive/exclusive_dataset_inventory.csv
Purpose	Documents the KJV 1611 exclusive corpus inventory used to preserve the 15 non-aligned works and 5,705 units for future nearest-neighbor and cross corpus positioning analysis.
Manuscript use	Supports RQ4, H4, Study Scope, and Future Work.

Appendix G: Reproducibility Manifest and Repository Release Metadata

This appendix identifies the public repository, release metadata, manifest files, notebooks, figures, architecture documentation, and manuscript materials associated with the study. It serves as the central reproducibility index for locating the artifact groups referenced throughout Appendices A–F. The released repository contains generated study outputs and analysis materials needed to inspect, validate, and reproduce the reported tables and figures. The proprietary InterLink Research Platform source code is not publicly released. All repository paths listed in Appendices A–F are relative to the repository root identified in the table below.

Field	**Description/Value**
Repository Root	https://github.com/Saorsa-Ind/kjv-1611-1769-semantic-stability-study
Repository Release Tag	v1.0.1
Run Name	drift_20260610T223724Z
Primary Manifest Files	MANIFEST.md, DATA_AVAILABILITY.md, manifest/manifest_run.json, manifest/manifest_outputs.json
Citation Metadata	CITATION.cff
Validation/Analysis Notebook	notebooks/Study_EDA_Validation_Figs_TableData.ipynb
Architecture Documentation	architecture/
Generated Figures	figures/
Manuscript Files	manuscript/
InterLink Source Code Status	Proprietary platform source code is not publicly released.